

Towards one Integrated System Vision

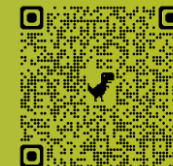


Lecture ULB – Insights from Fluxys ‘Integration North Sea Model’

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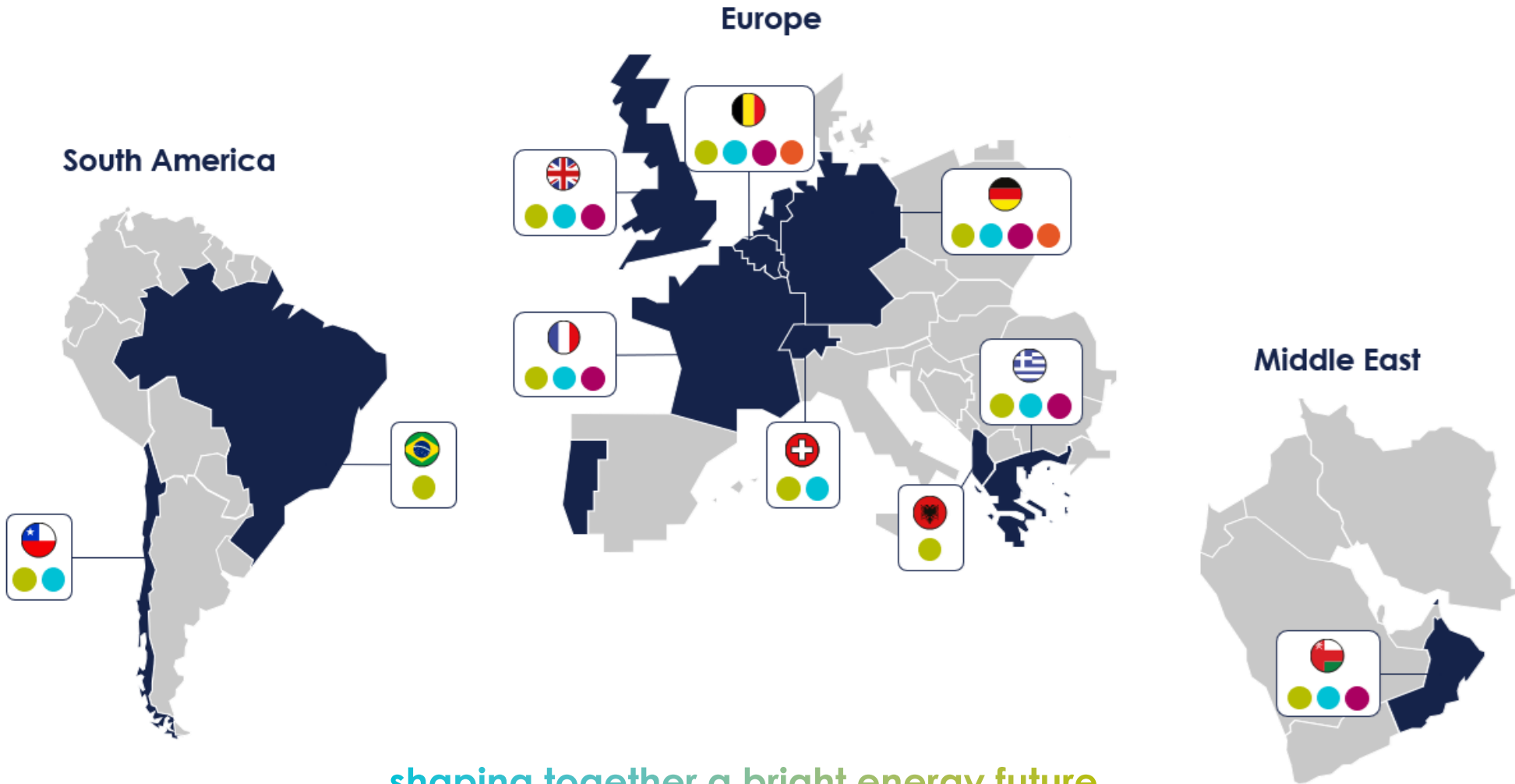
12/03/2026

Our studies can be
found on [our website](#)



Who is Fluxys?

Multi-molecule infrastructure partner headquartered in Belgium



shaping together a bright energy future

CH₄

H₂

CO₂

Elec

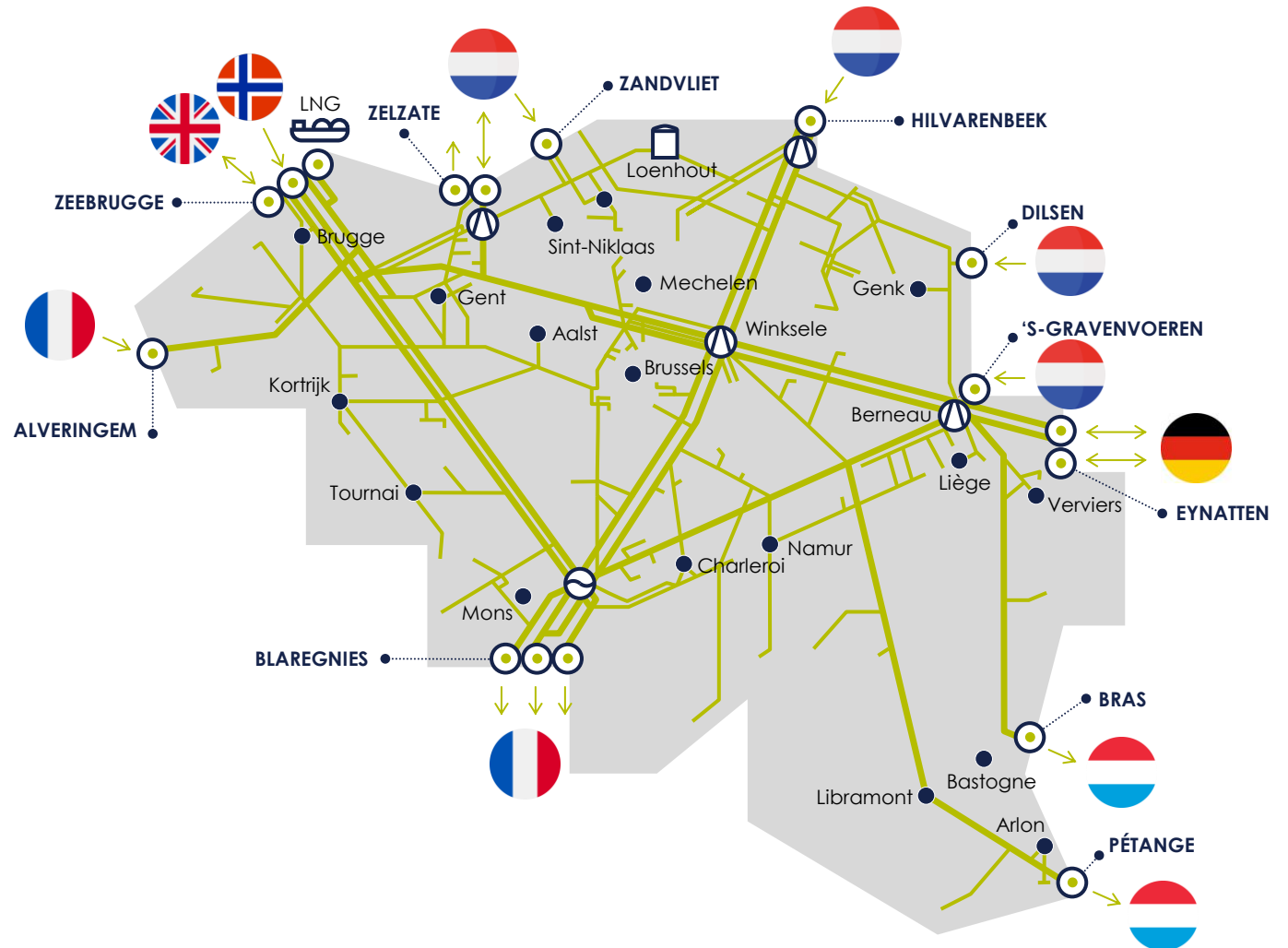


Who is Fluxys?

High pressure (15-80 bars) transmission infrastructure in Belgium

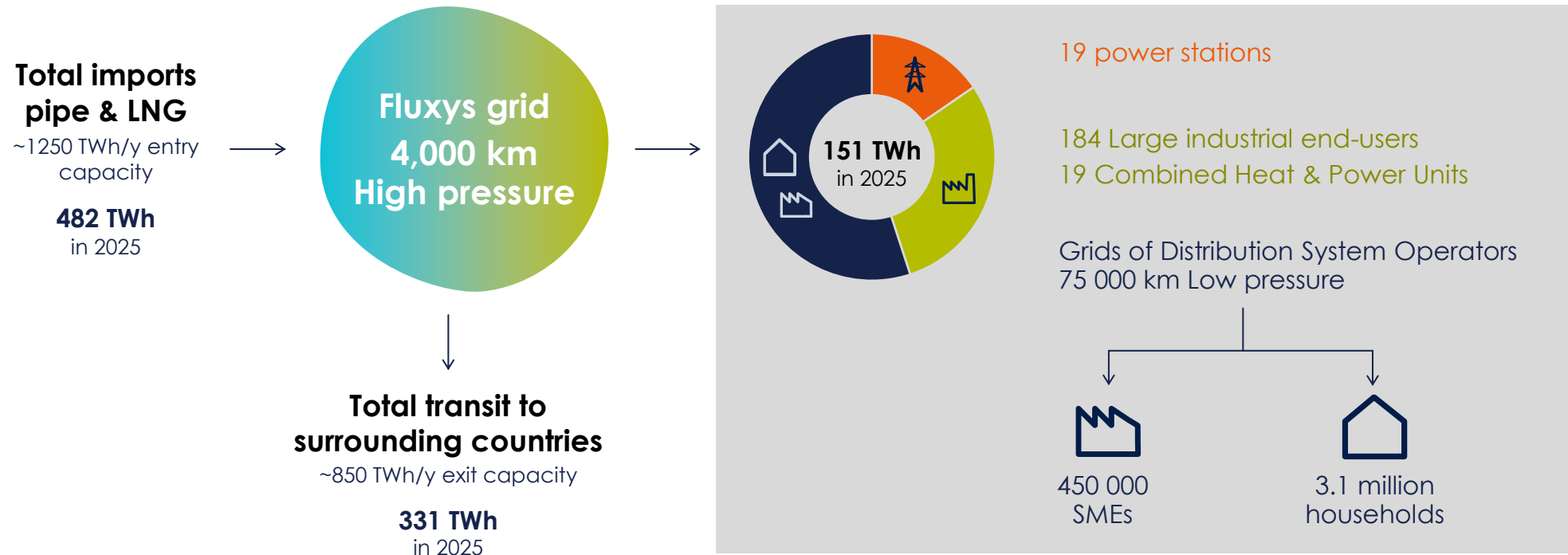
**4,000 km pipelines in
417 municipalities**

-  Physical interconnection points
-  LNG terminal
-  Compressor stations
-  Blending stations
-  Storage



Who is Fluxys?

Fluxys role on the Belgian gas market

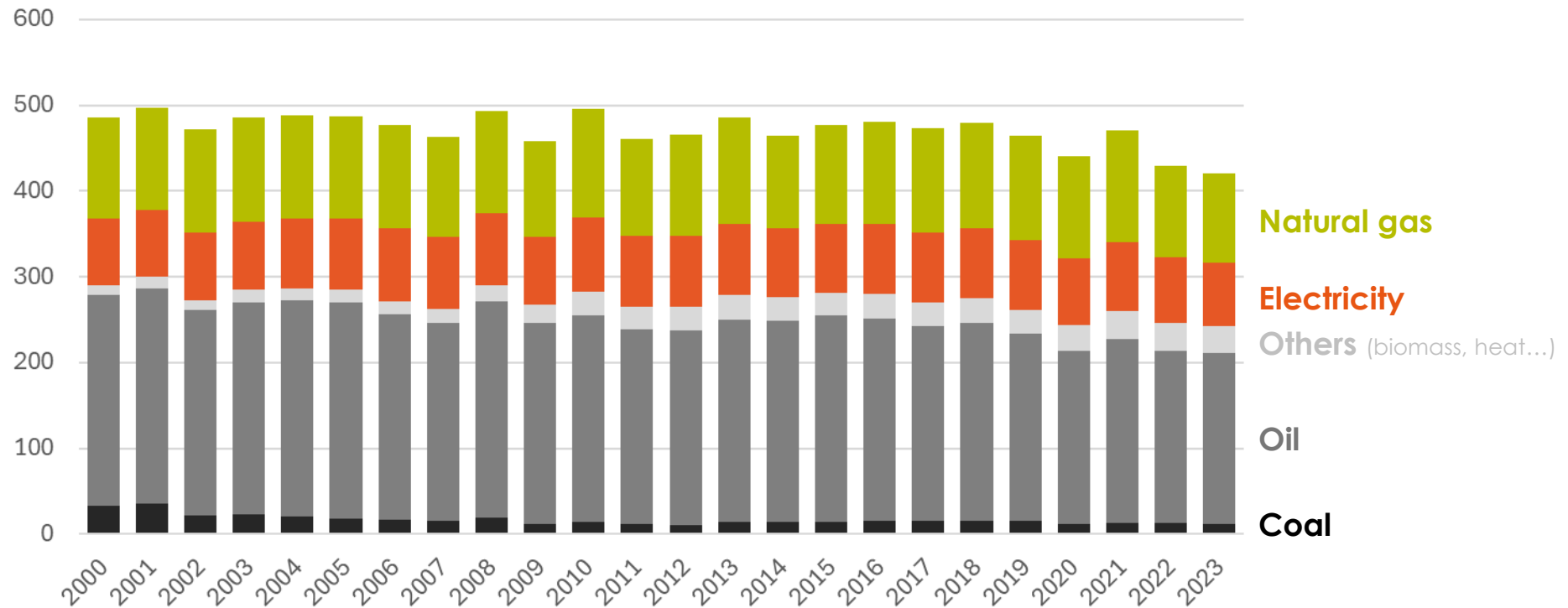


Natural gas is part of the total energy demand

Final energy demand for Belgium



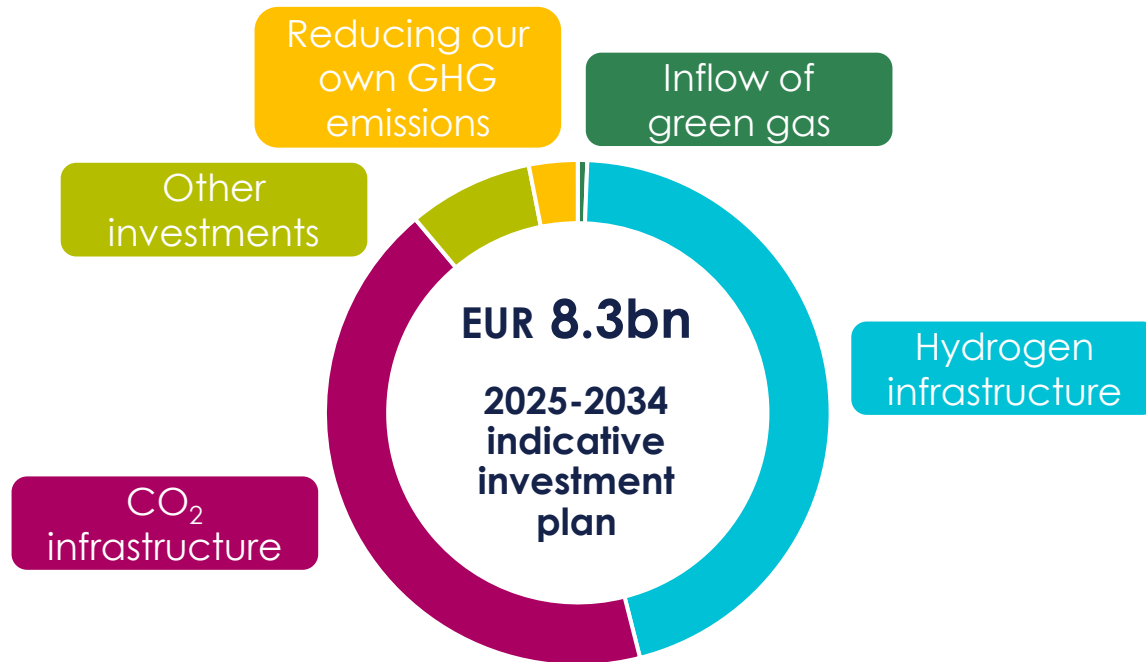
Final energy demand (TWh)



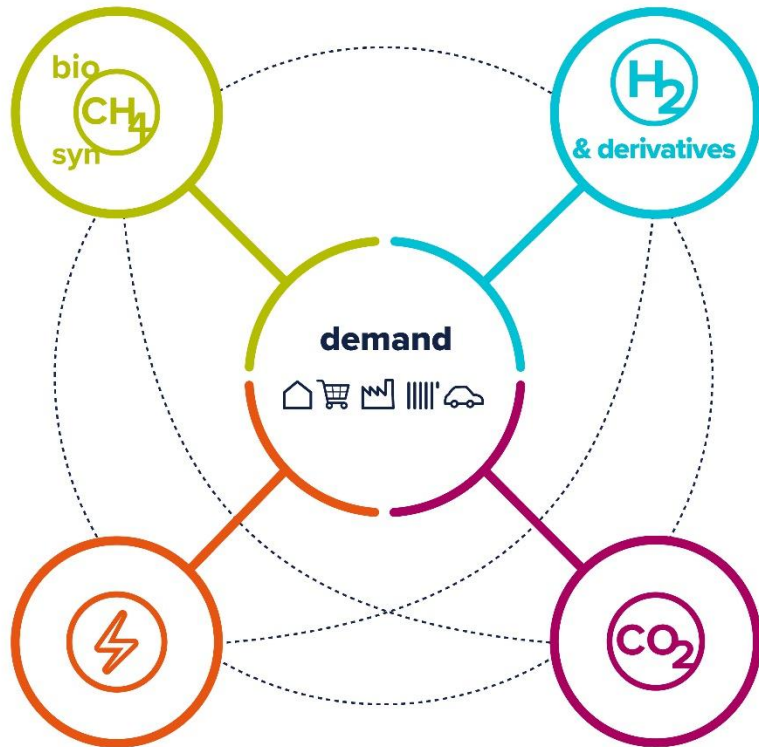
Source: IEA (<https://www.iea.org/countries/belgium/energy-mix>)



90% of Fluxys Belgium investment plan earmarked for hydrogen & CO₂ infrastructure and reducing its own carbon footprint



Towards an integrated approach to the entire energy system: Fluxys has built a multi-energy simulation model



all interactions
between electricity,
methane, hydrogen
and CO₂ on an **hourly**
basis



energy systems of
all countries
bordering the
North Sea



achieve **carbon-**
neutrality at the
lowest cost with
energy demand
served at any time

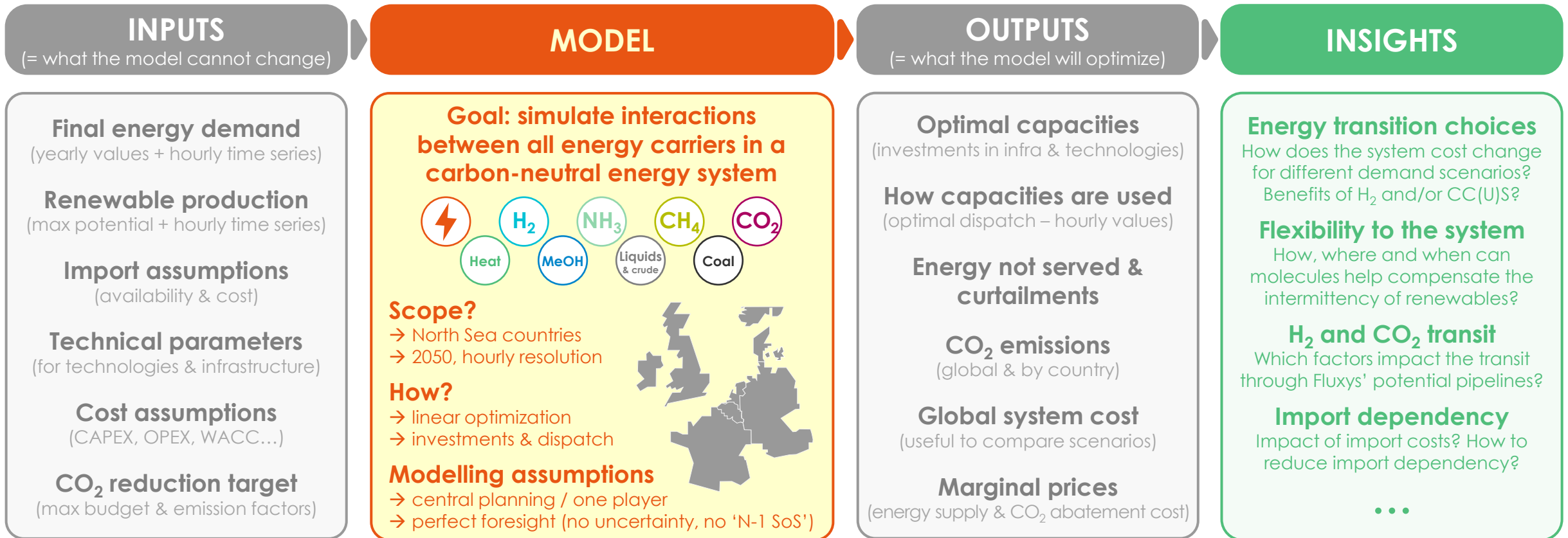


Informed decision-making
on infrastructure planning across sectors

1. THE MODEL

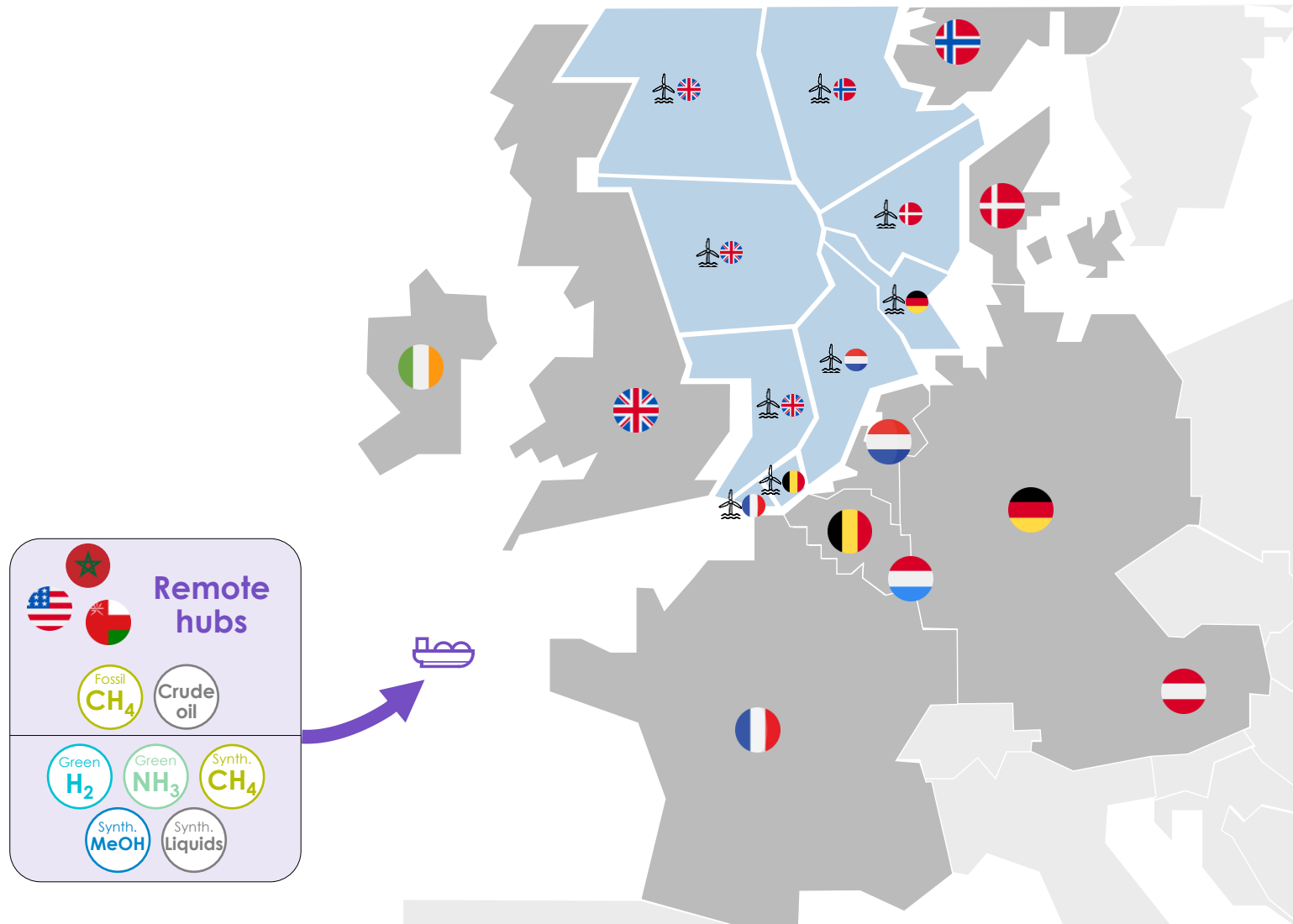


The model aims to identify the optimal multi-energy system that can achieve carbon neutrality by 2050



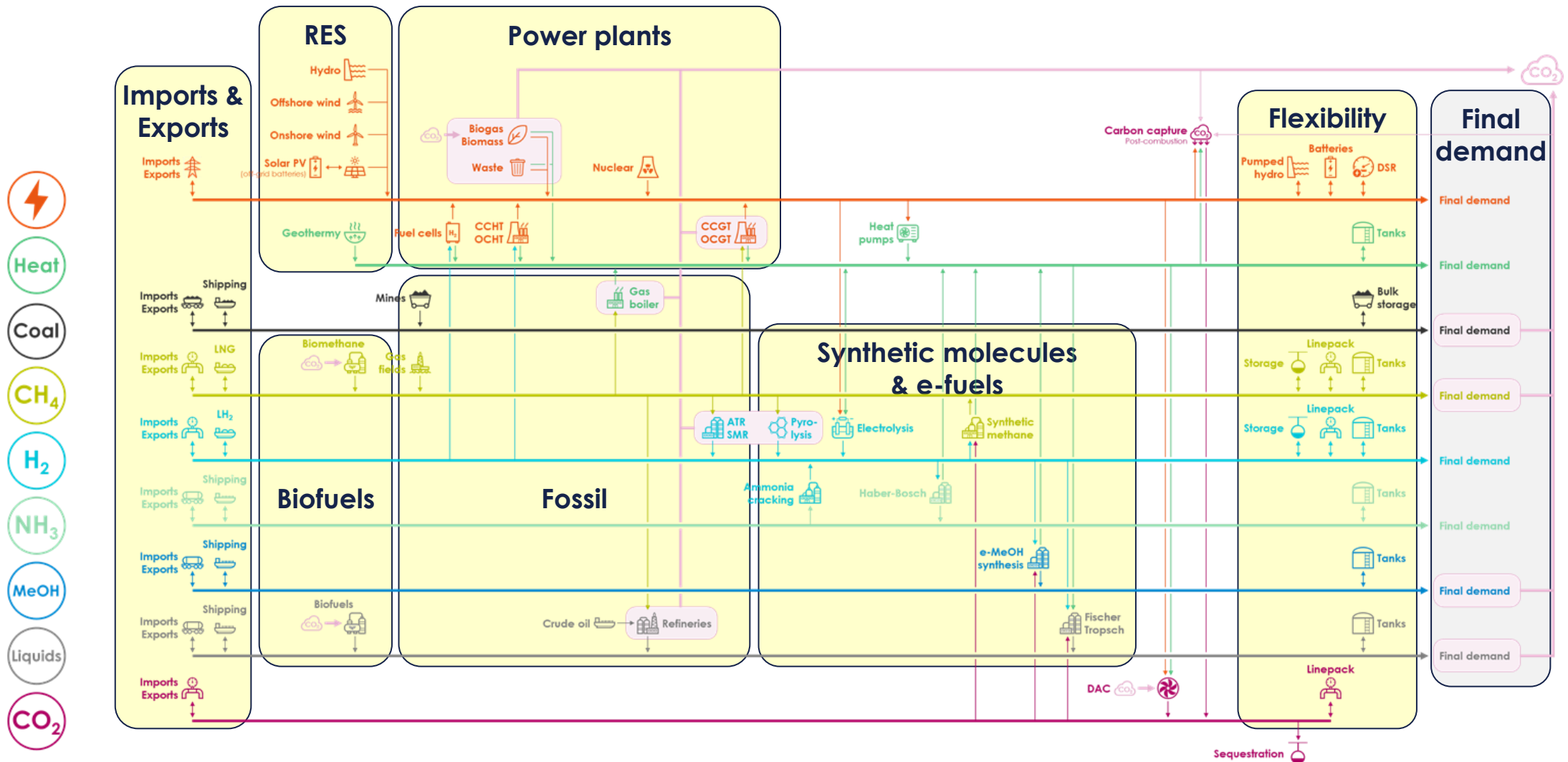
Integrated energy system with 9 carriers

Geographical scope



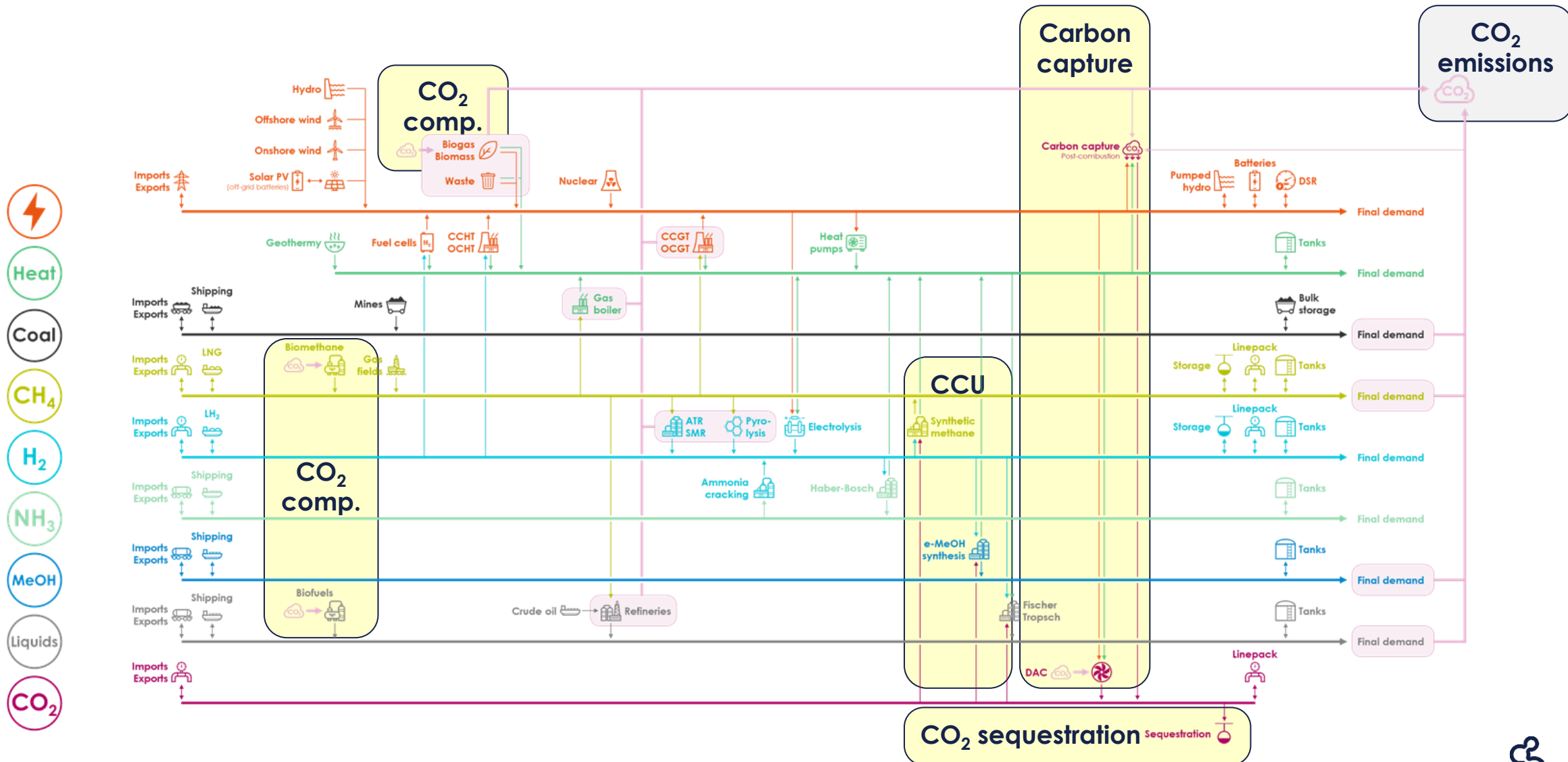
Integrated energy system with 9 carriers

Carriers and technologies



Integrated energy system with 9 carriers

CO₂ emissions can be compensated or captured

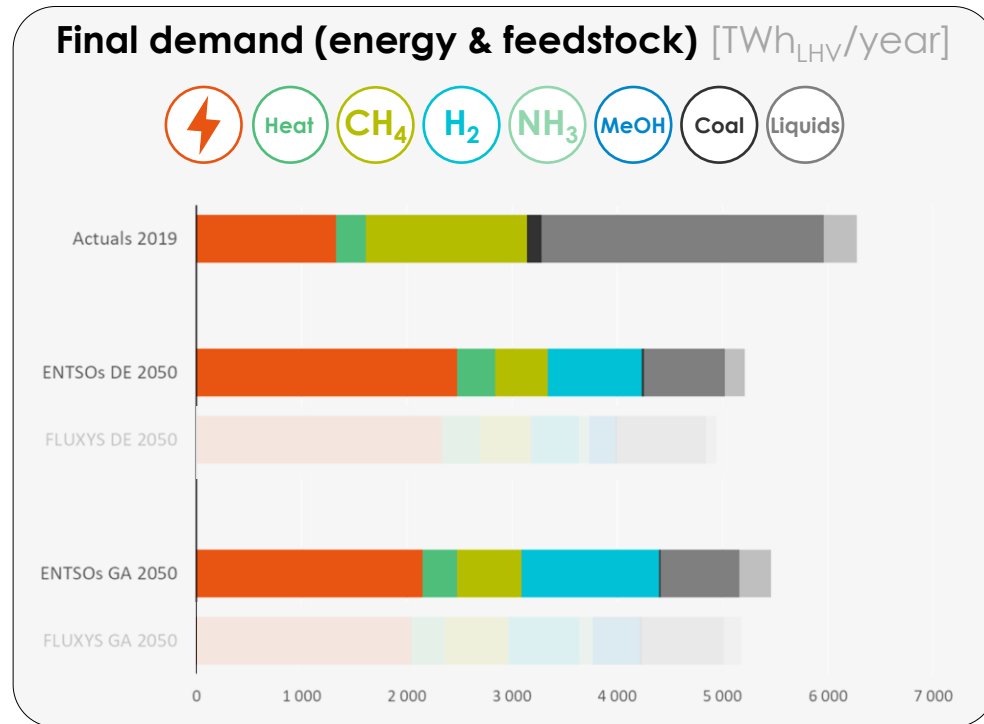


2. SCENARIOS & ASSUMPTIONS



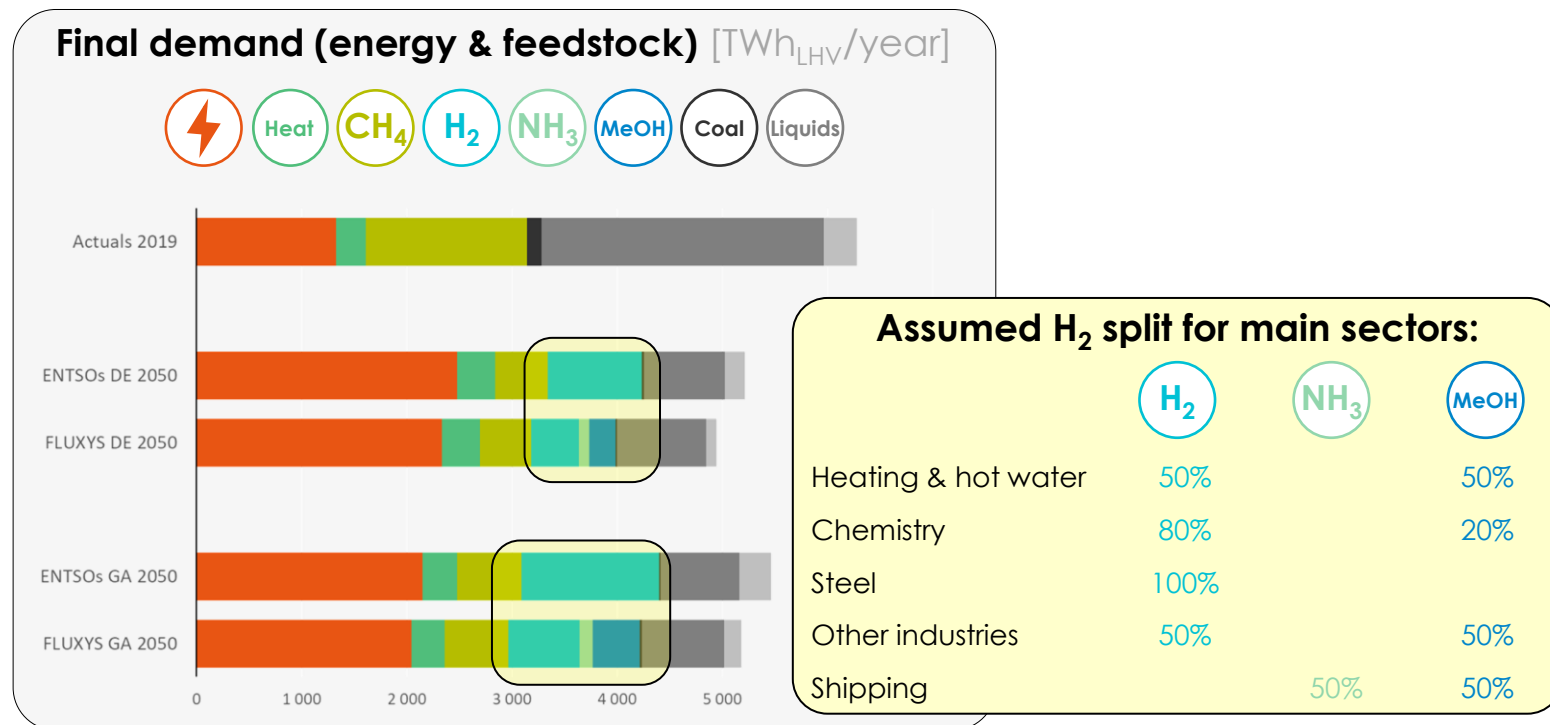
Starting position is a final demand scenario

Values from Global Ambition and Distributed Energy scenarios (ENTSOs' TYNDP 2024) were taken as final demand for 2050



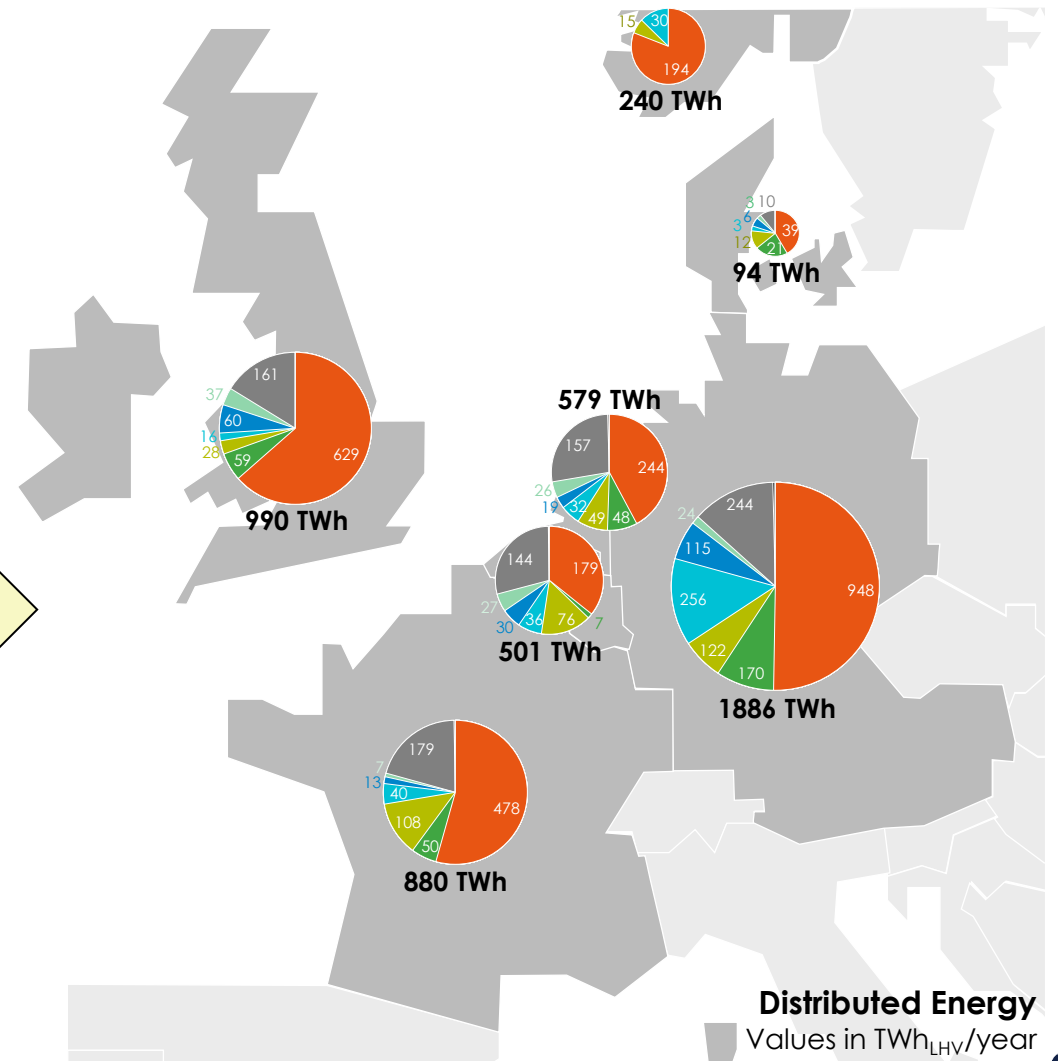
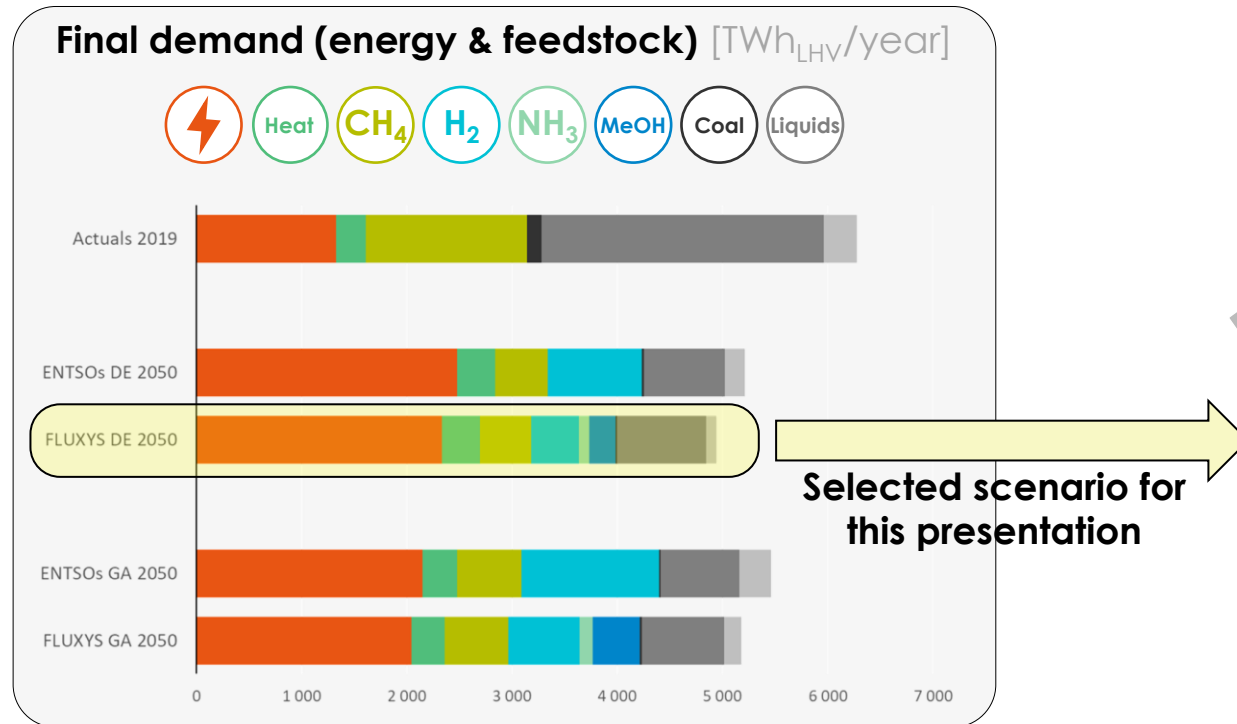
Starting position is a final demand scenario

Values from Global Ambition and Distributed Energy scenarios (ENTSOs' TYNDP 2024) were taken as final demand for 2050... and slightly modified for hydrogen derivatives



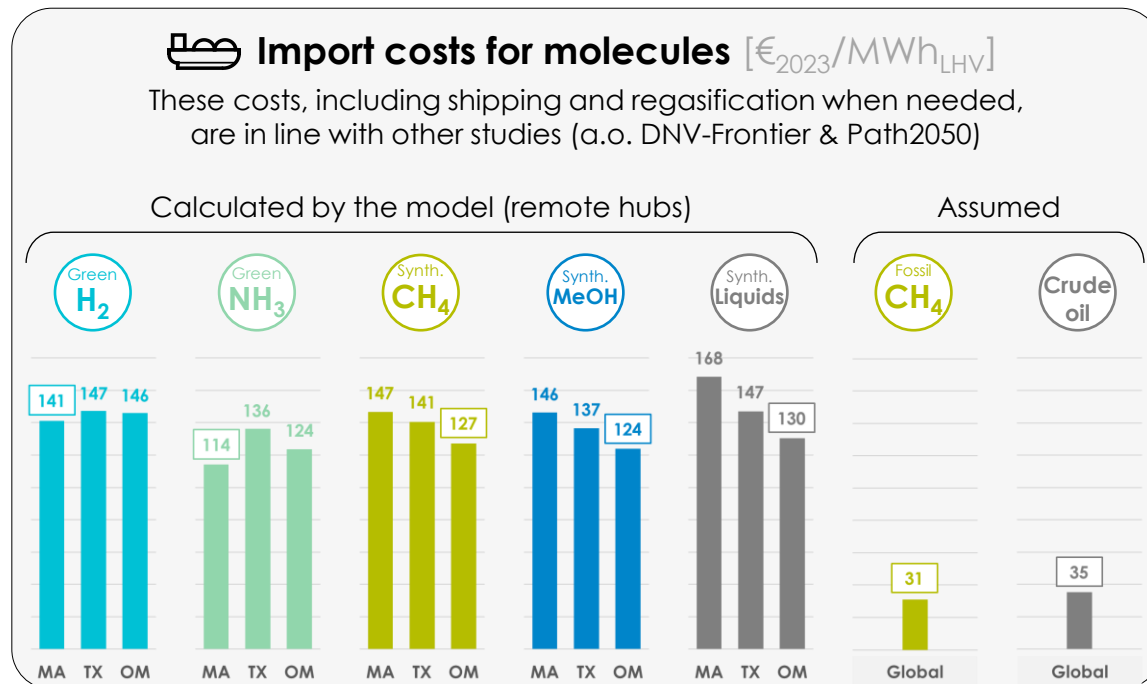
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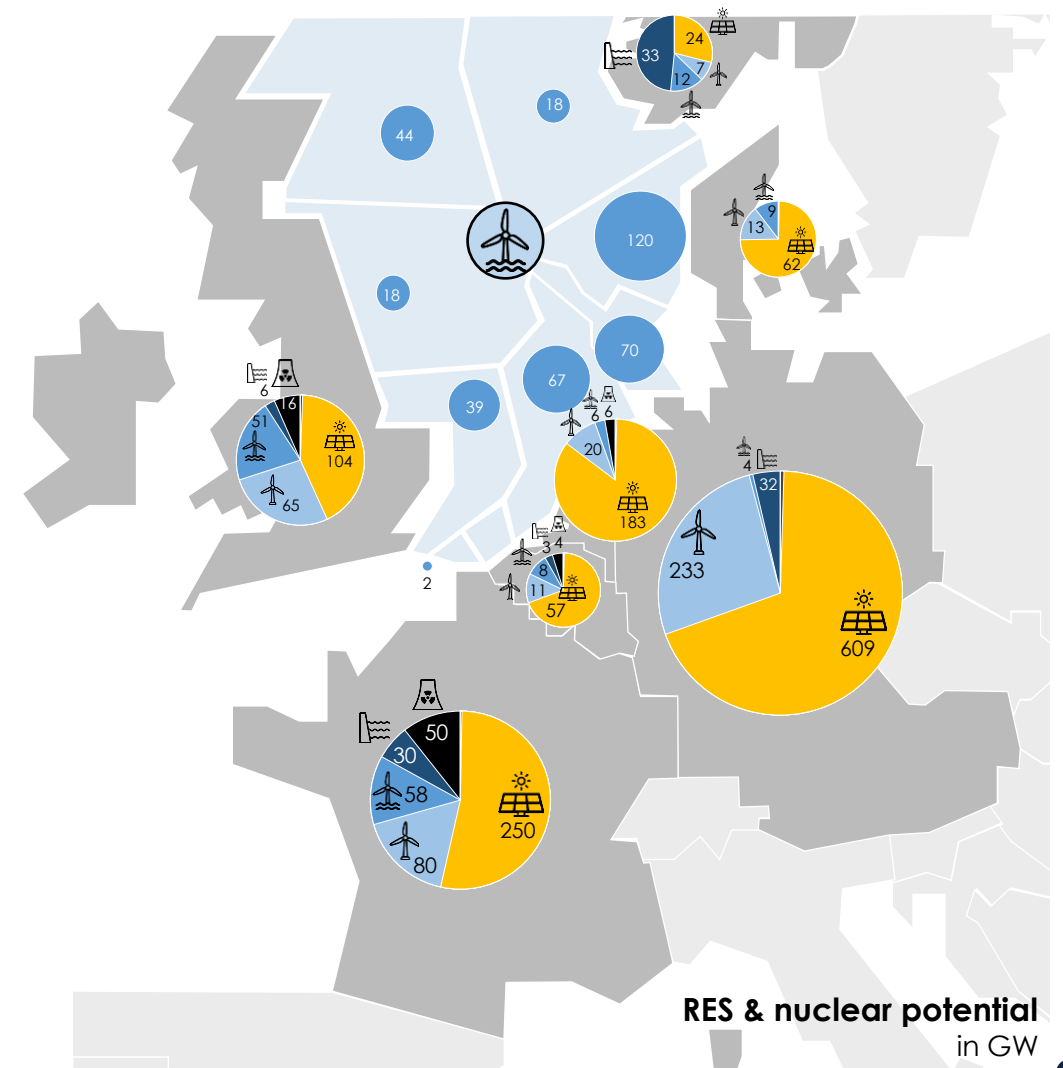


Potentials for RES and nuclear vary across countries

Maximal potentials for renewable electricity in 2050
come from ENTSOs' TYNDP 2024



Other cost & technical assumptions come mainly from the Danish Energy Agency
(<https://ens.dk/en/our-services/technology-catalogues>)



3. KEY INSIGHTS

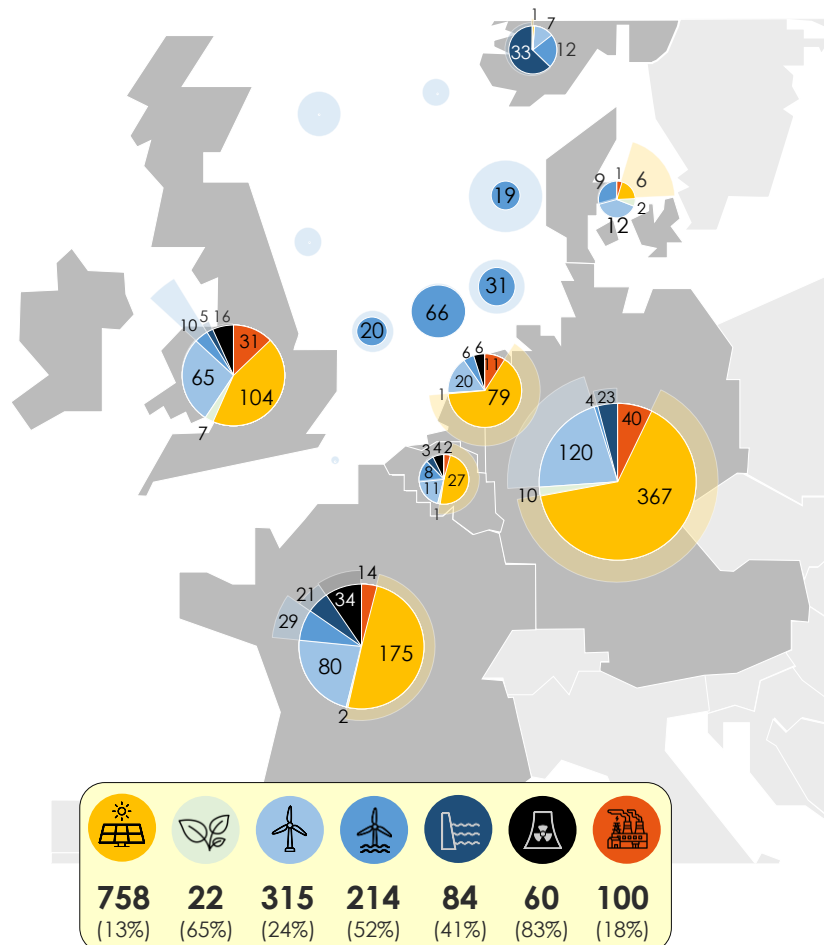


1. Renewables produce almost all electricity (1/3)

Wind and solar PV are massively deployed and complemented by hydro, nuclear and biomass to supply final demand and green hydrogen production

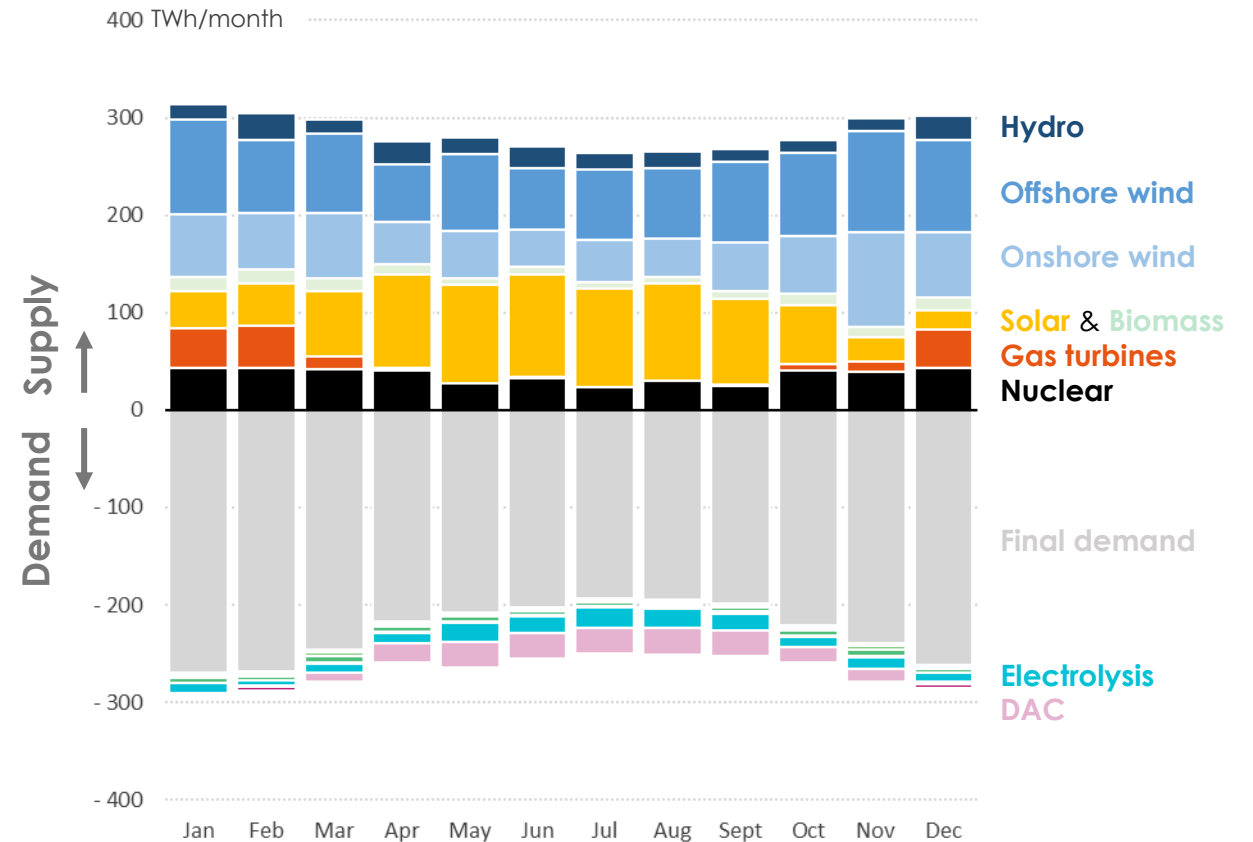
Electricity generation capacity

(Installed capacities in GW and load factors in %)



Electricity balance

(Distributed Energy scenario)

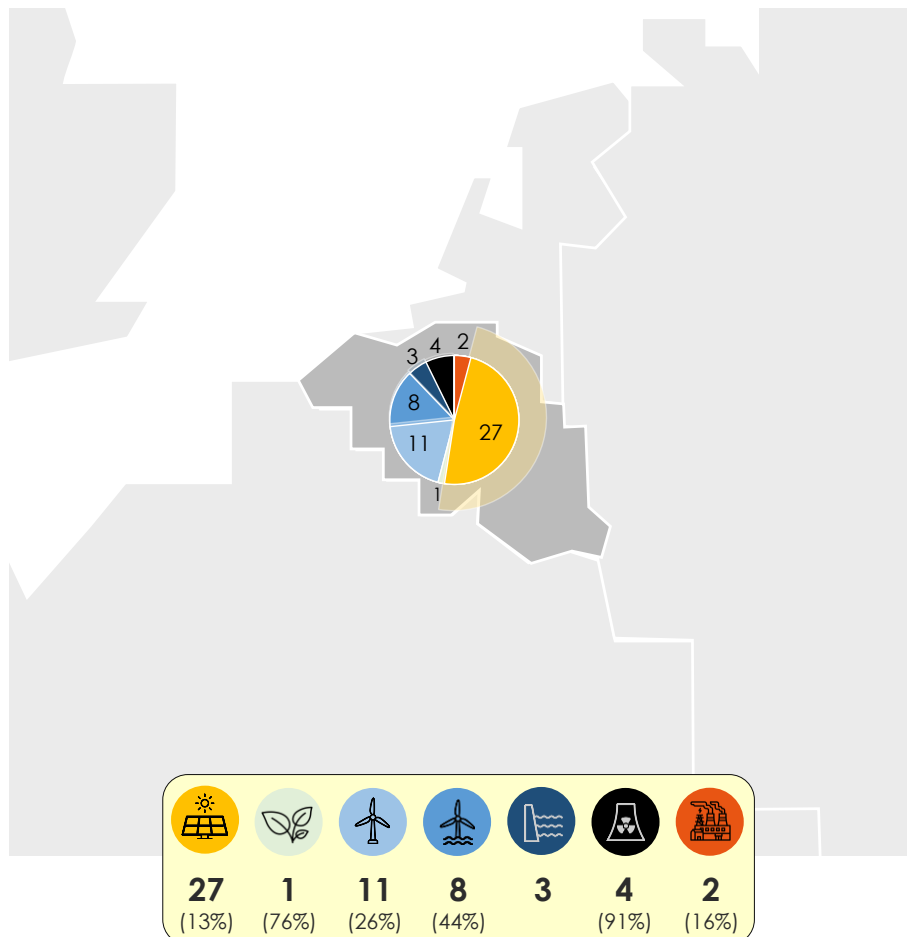


1. Renewables produce almost all electricity (2/3)

In Belgium, wind and solar PV are massively deployed, but nuclear and strong imports are necessary on top of it to balance final demand

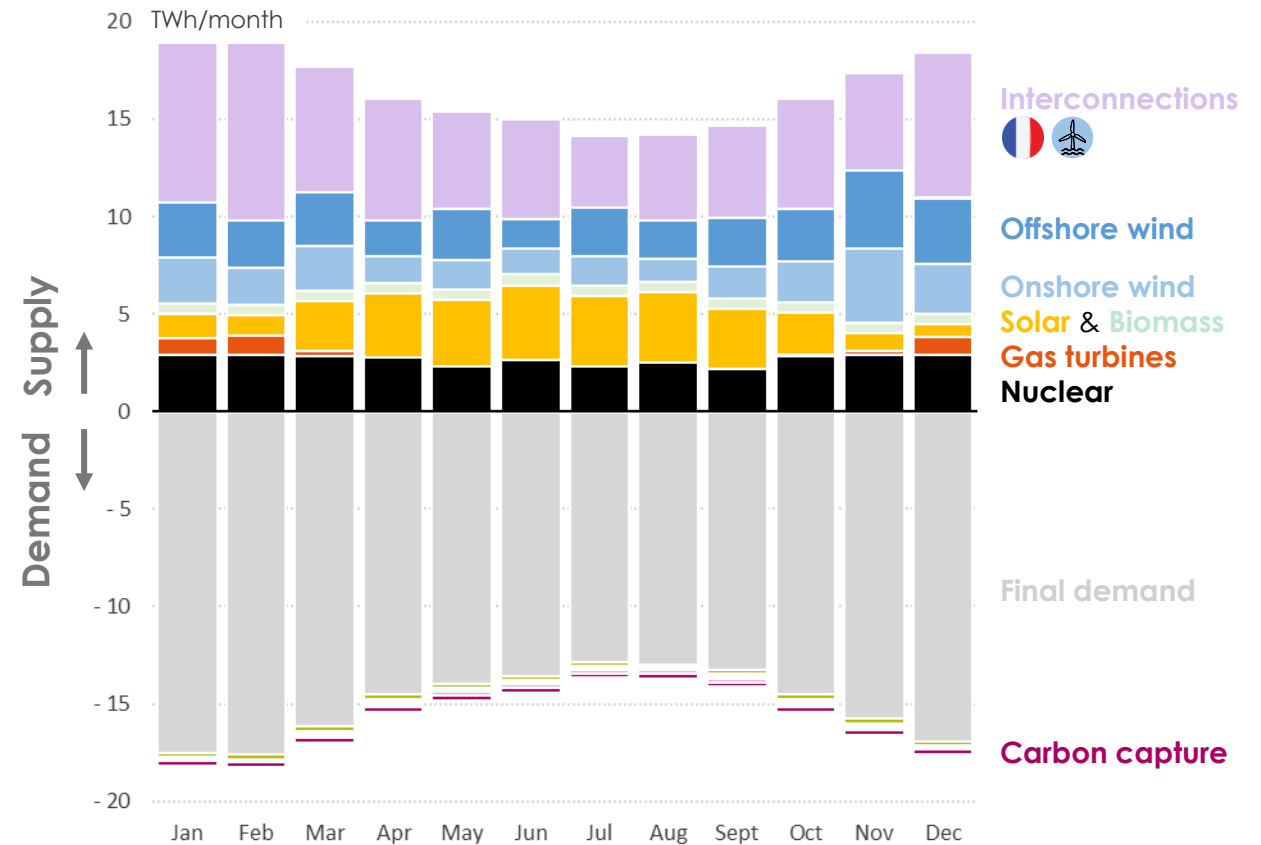
Electricity generation capacity

(Installed capacities in GW and load factors in %)



Electricity balance

(Distributed Energy scenario)



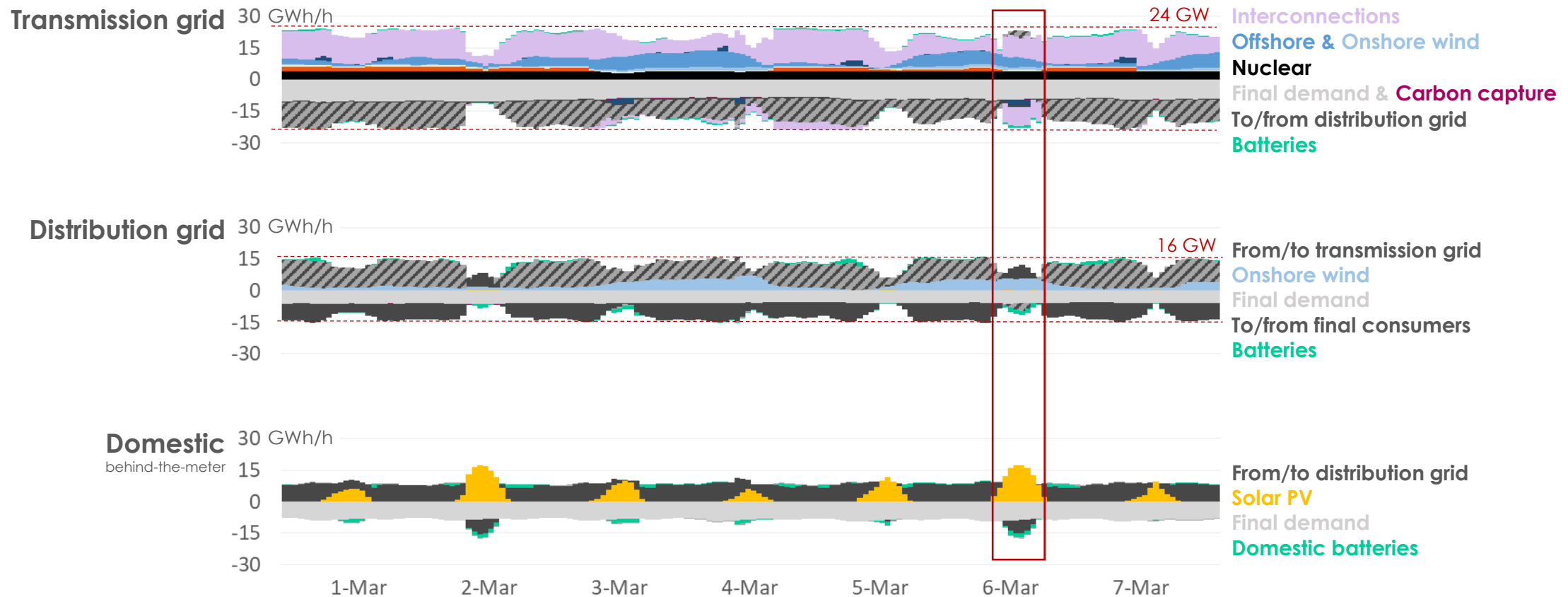
1. Renewables produce almost all electricity (3/3)

In Belgium, the full potential of solar PV is not deployed to limit the reinforcement of the transmission and distribution grids, but also to avoid excessive curtailment



Electricity balance in March

(Distributed Energy scenario)



2. Gas turbines necessary in winter

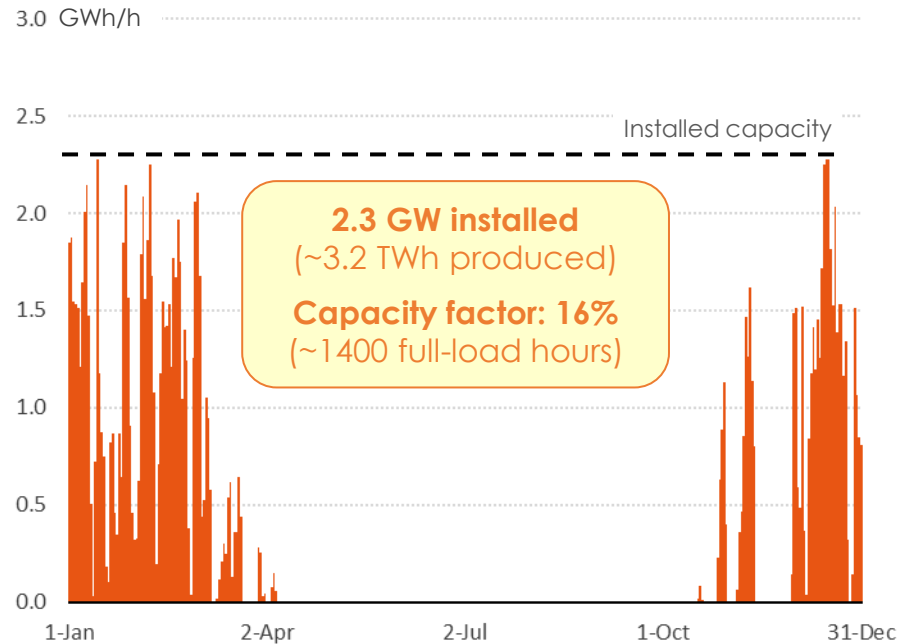
100 GW of dispatchable gas-to-power capacity is installed to secure electricity supply during winter (2.3 GW are located in Belgium)

(currently 85 GW of gas-fired power plants & 34 GW of coal-fired power plants – source ENTSO-e & Global Energy Monitor)



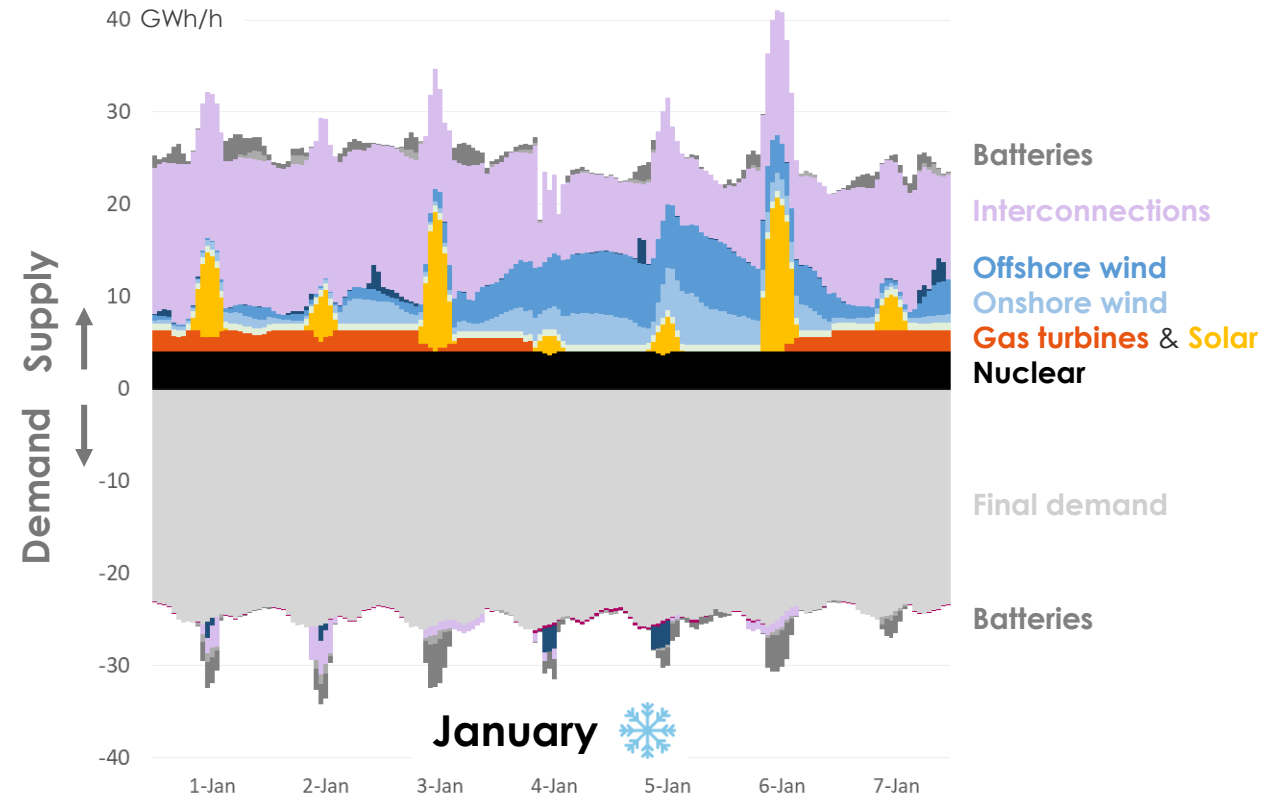
Gas turbines usage

(Distributed Energy scenario)



Electricity balance

(Distributed Energy scenario)

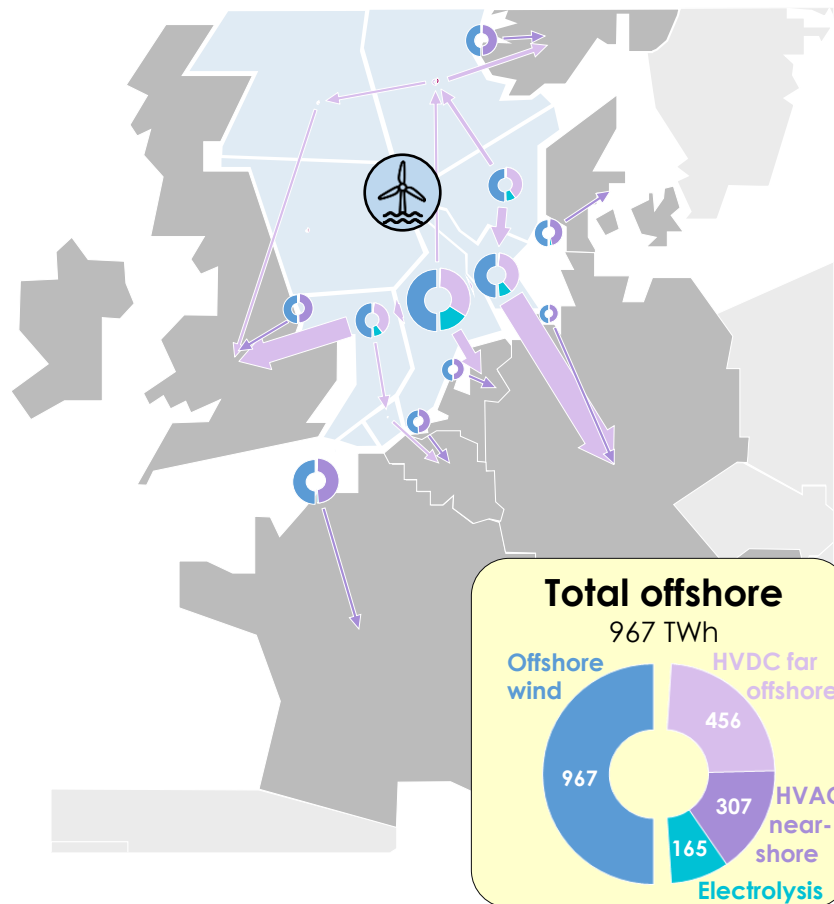


3. Offshore electrolysis helps deploying offshore wind

Excess wind production is converted to hydrogen to avoid curtailments and maximise wind capacity use – especially offshore

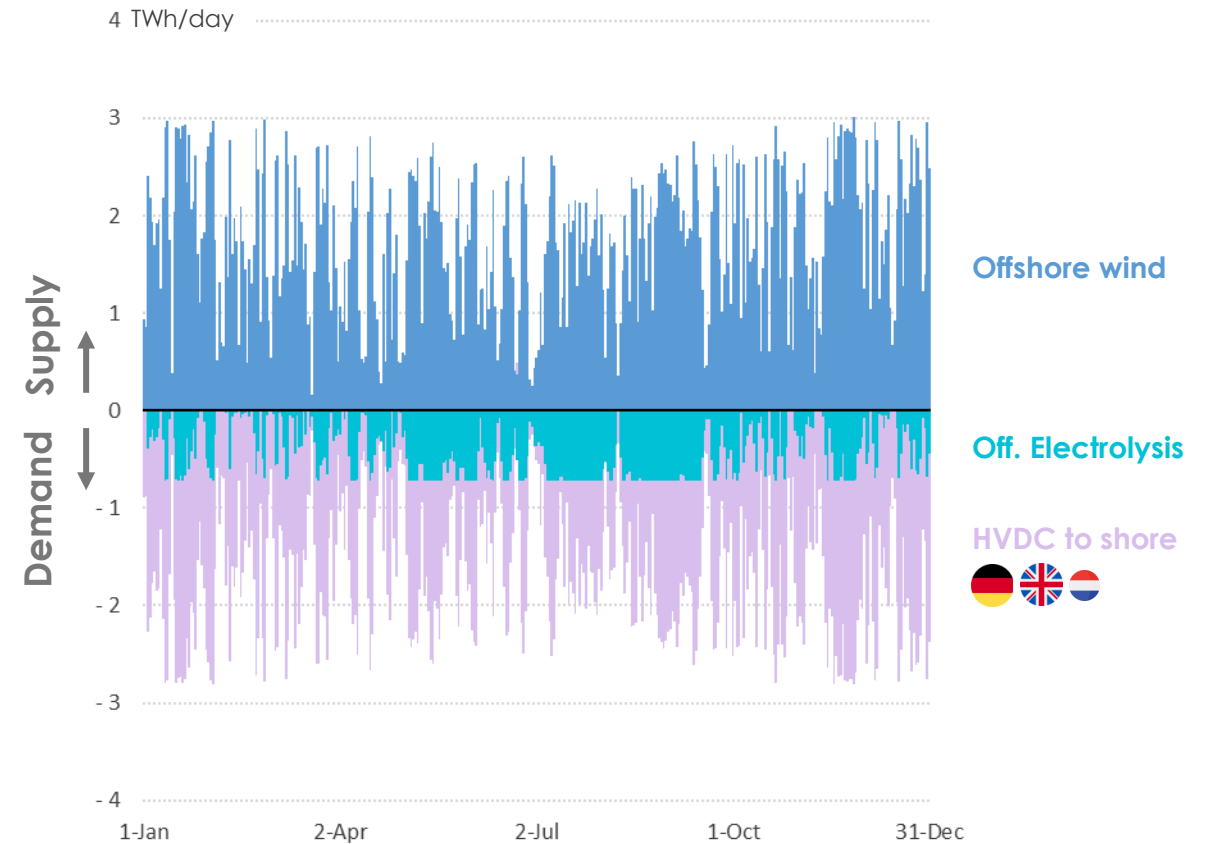
Electricity balance – Offshore

(Distributed Energy – Annual quantities in TWh)



Electricity balance – Far-Offshore

(Distributed Energy scenario)

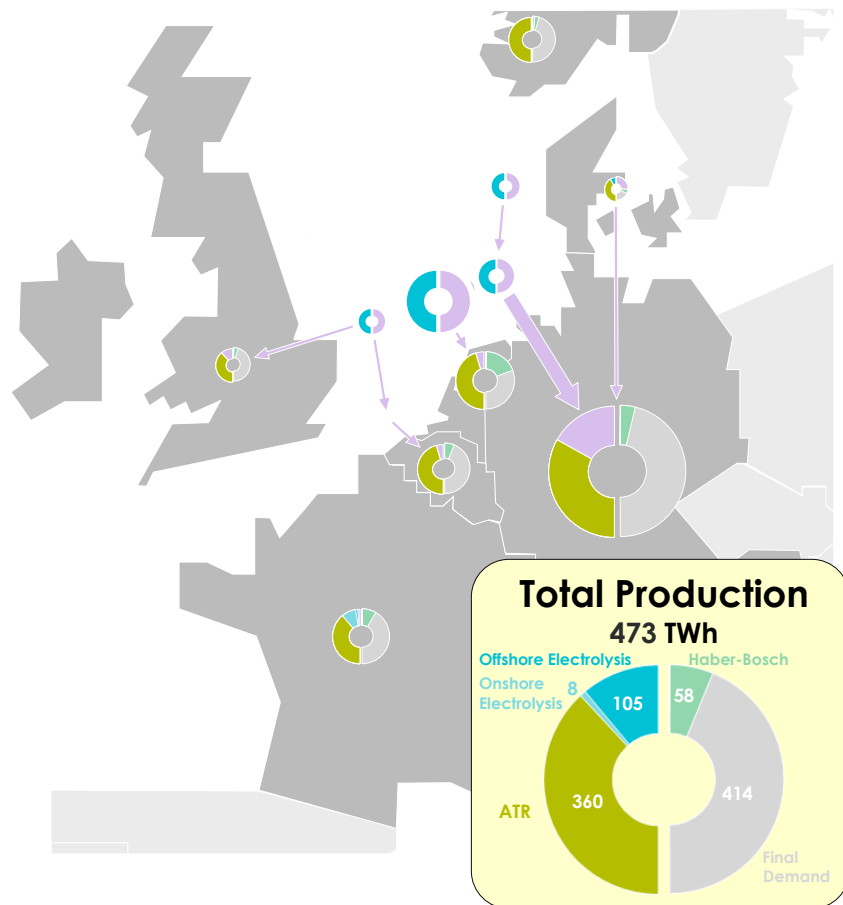


4. Gaseous hydrogen is fully produced within Europe

Hydrogen is not imported from overseas and can be produced by electrolysis of excess electricity (mainly offshore) and from methane by autothermal reforming (ATR)

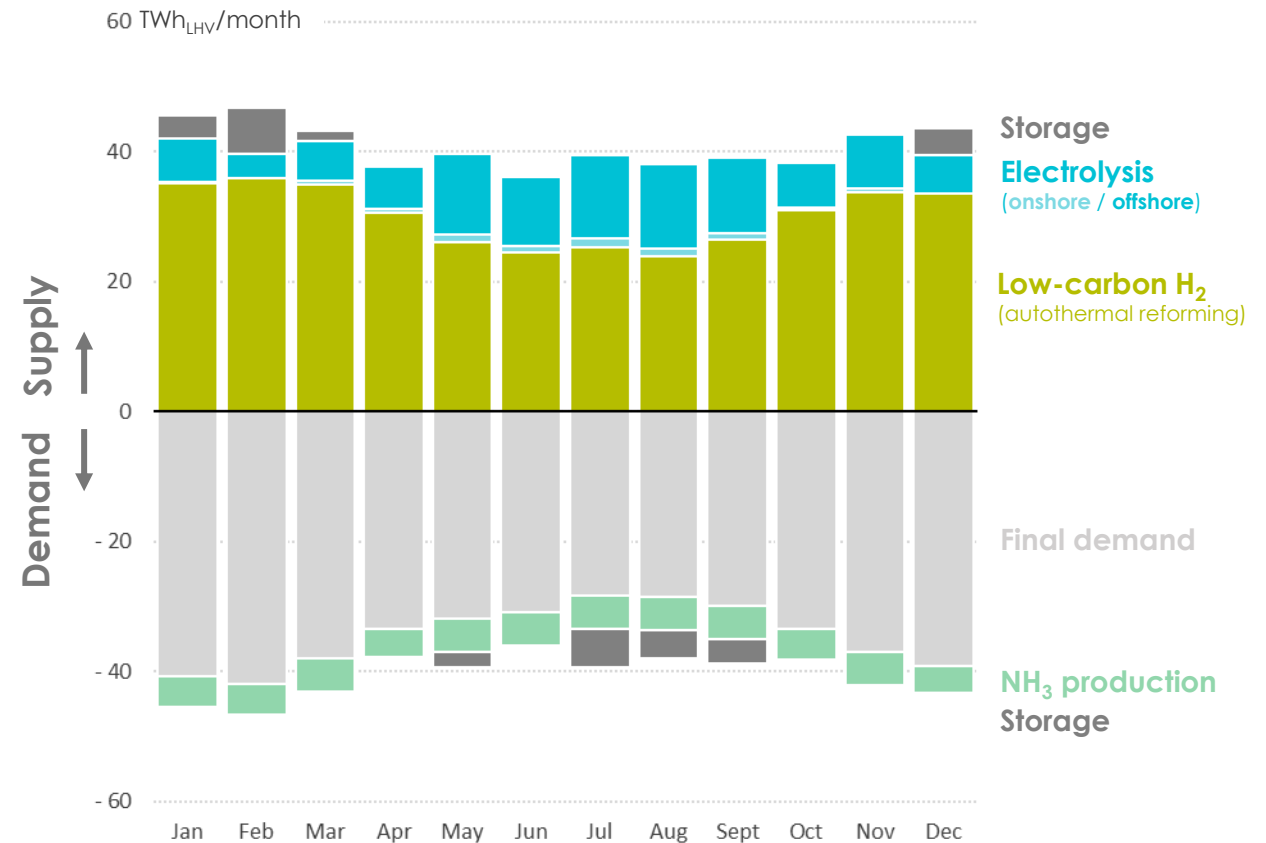
Hydrogen balance

(Distributed Energy – Annual quantities in TWh_{LHV})



Hydrogen balance

(Distributed Energy scenario)



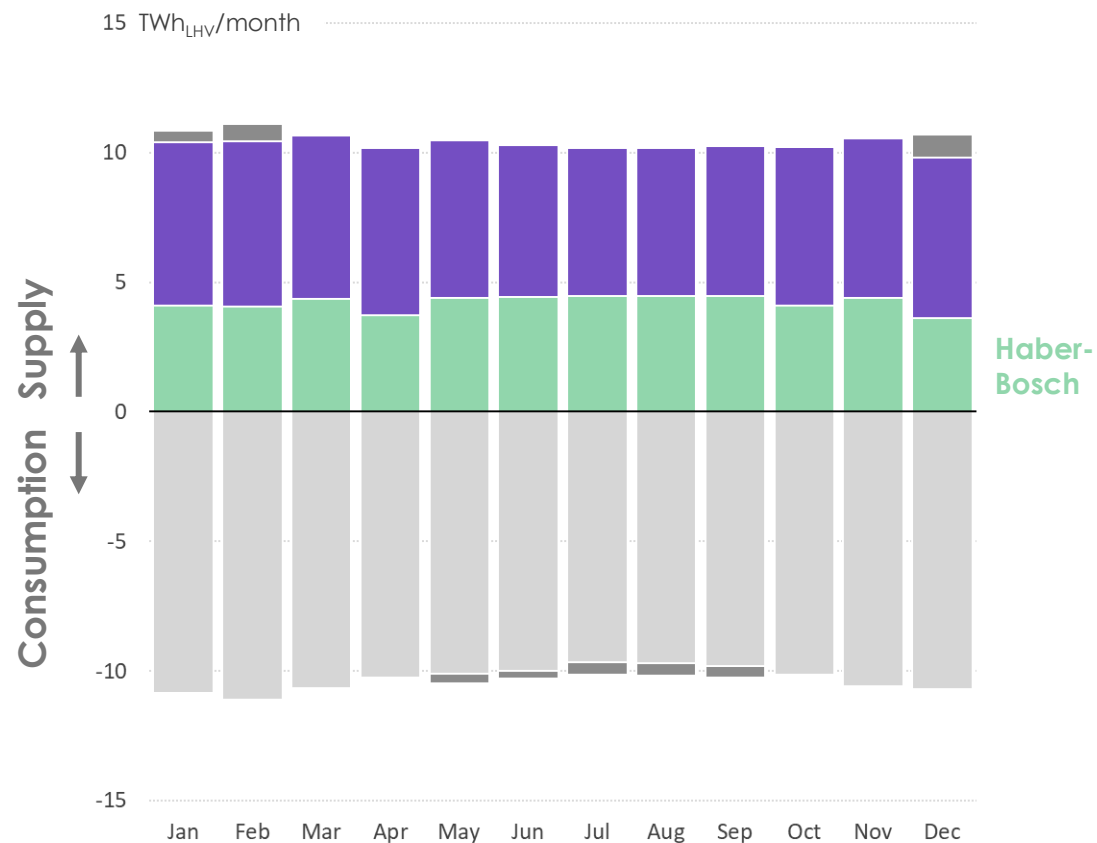
5. Hydrogen derivatives mostly imported from overseas

Only a small proportion of NH_3 is produced locally (Haber-Bosch process), as importing green ammonia and methanol from remote hubs is cheaper



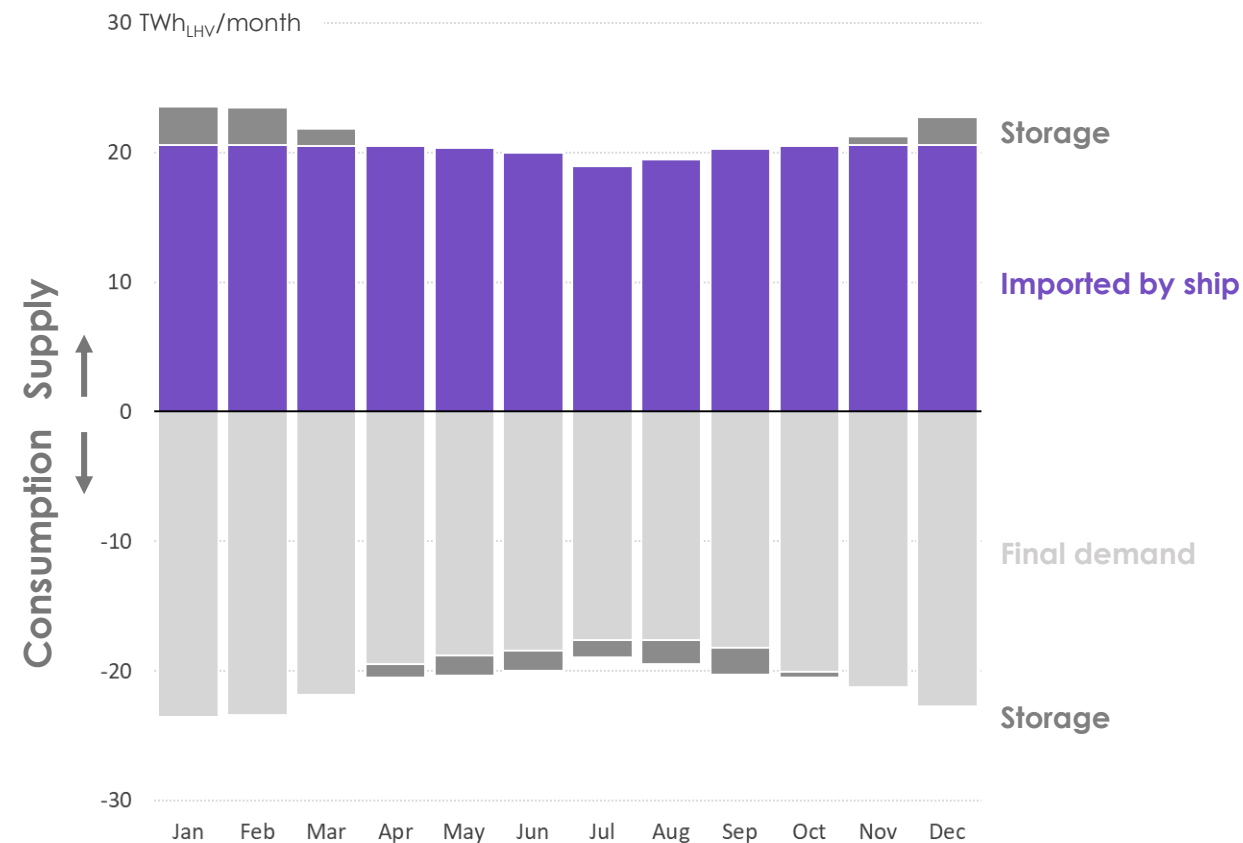
Ammonia balance

(Distributed Energy scenario)



Methanol balance

(Distributed Energy scenario)

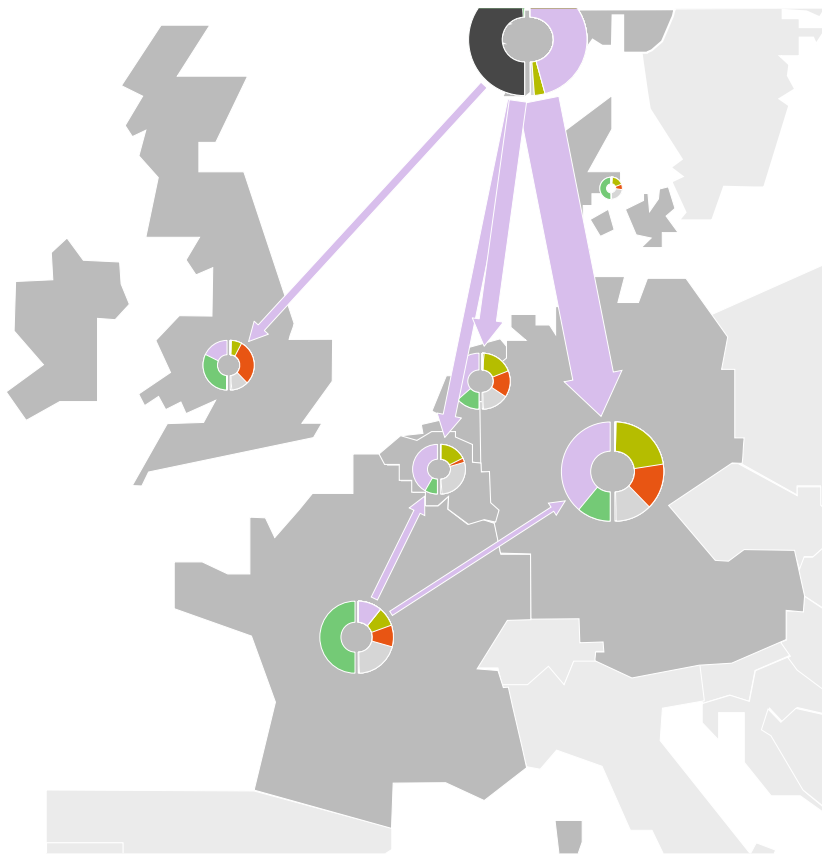


6. Biomethane is key

Aside conventional gas from Norway, biomethane production is maximized, enabling gas storages to be used for power demand during winter months

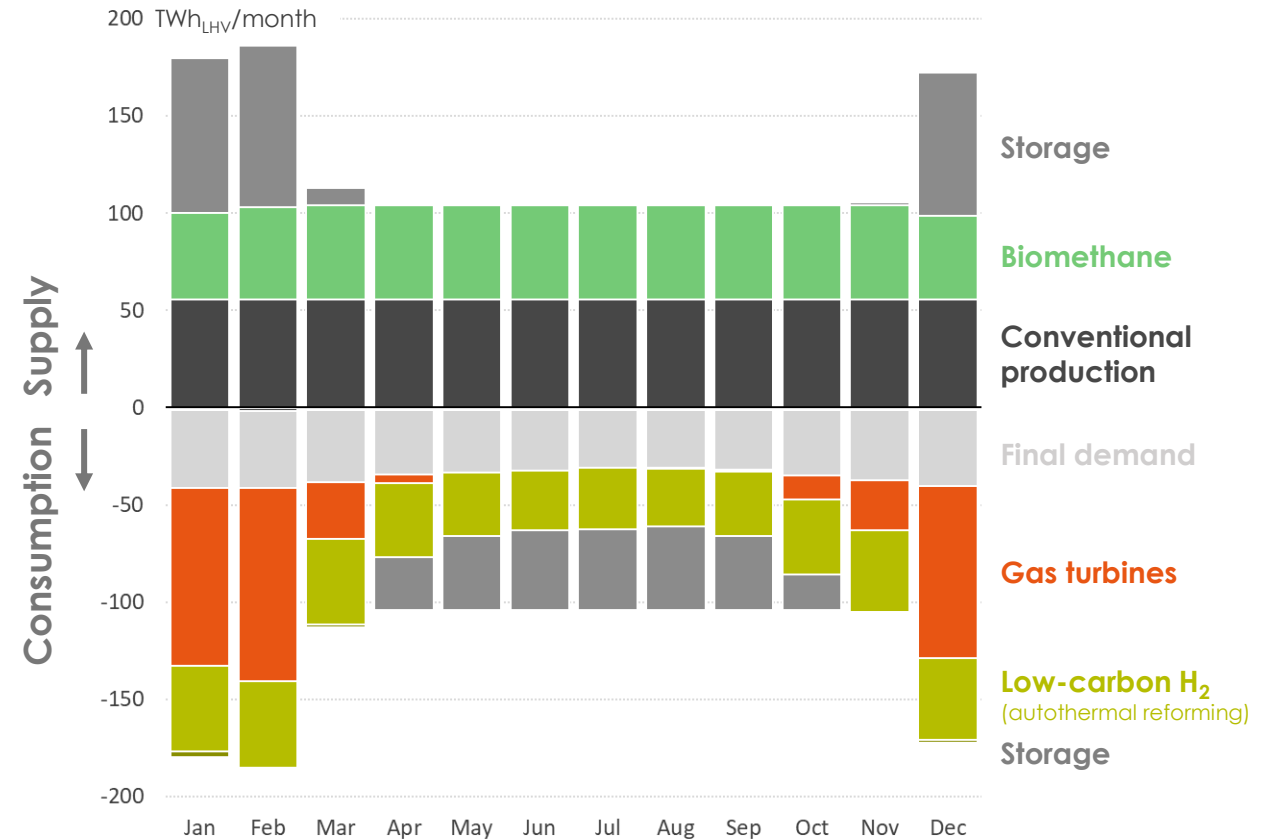
Methane supply

(Distributed Energy scenario)



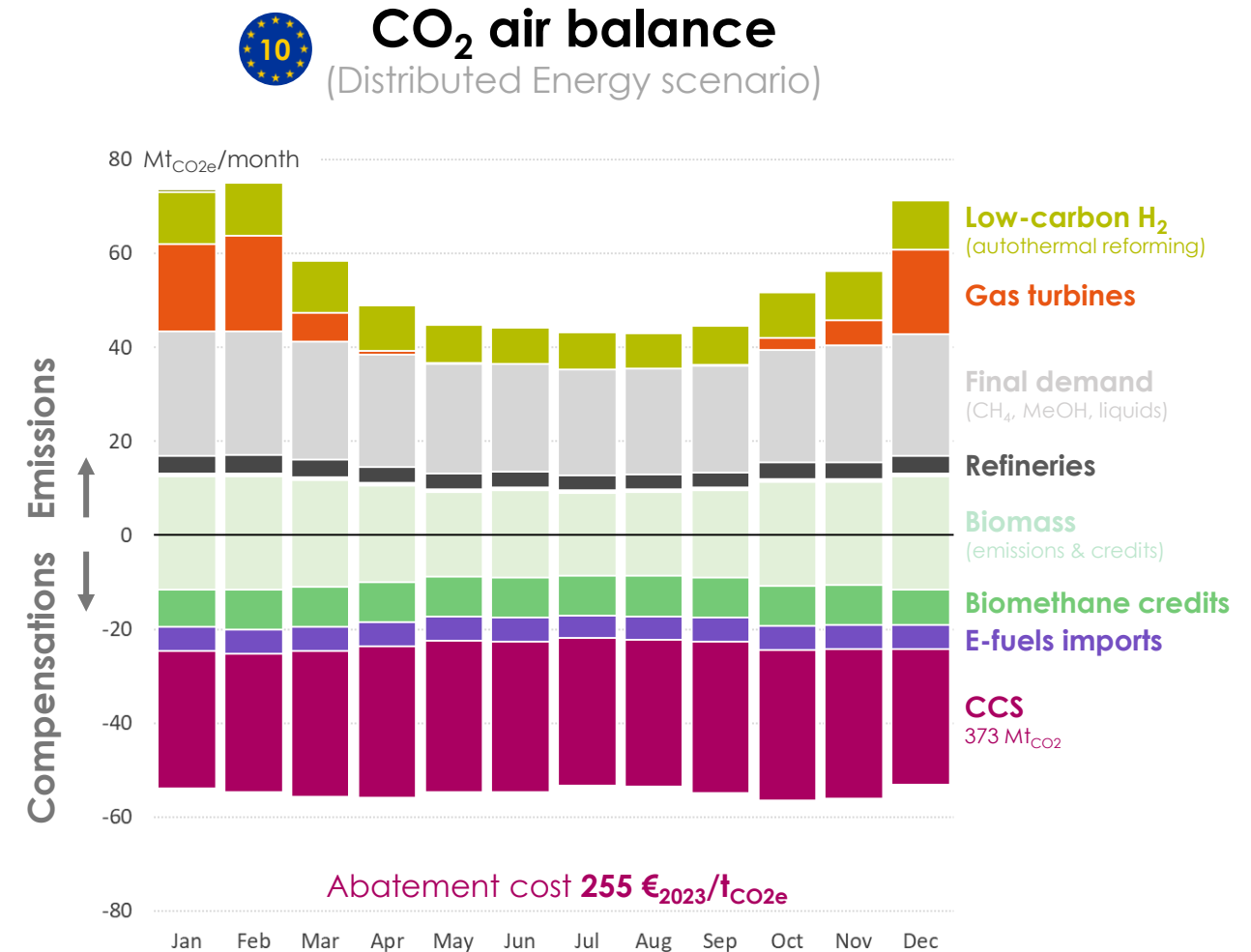
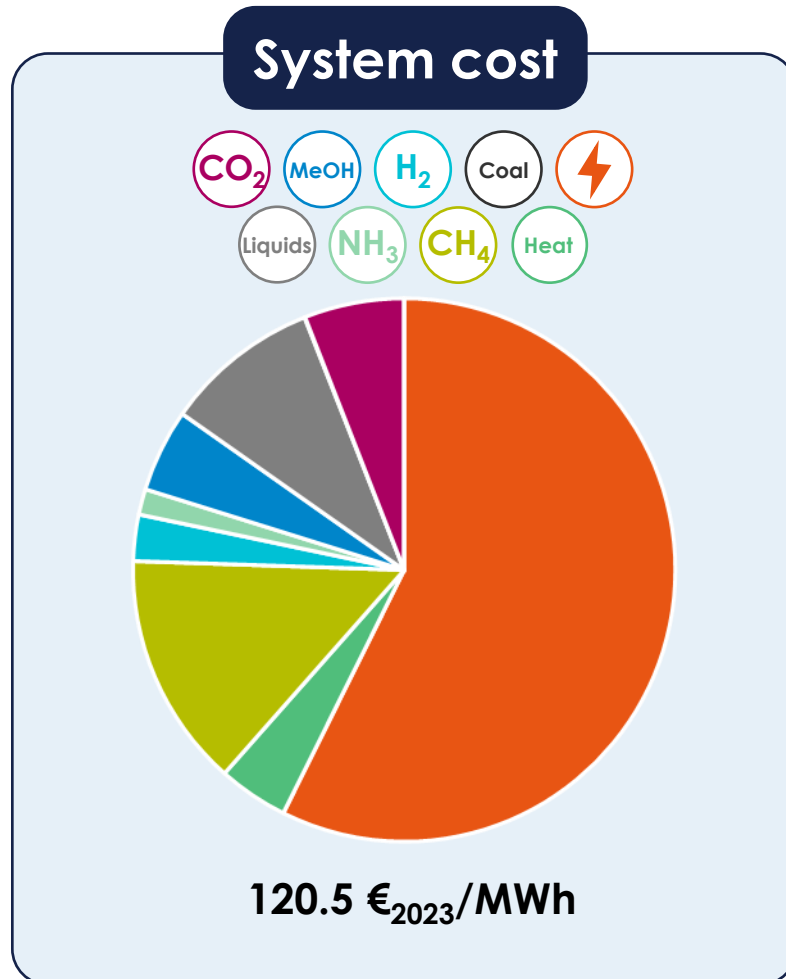
Methane balance

(Distributed Energy scenario)



7. Carbon capture is a no-brainer (1/2)

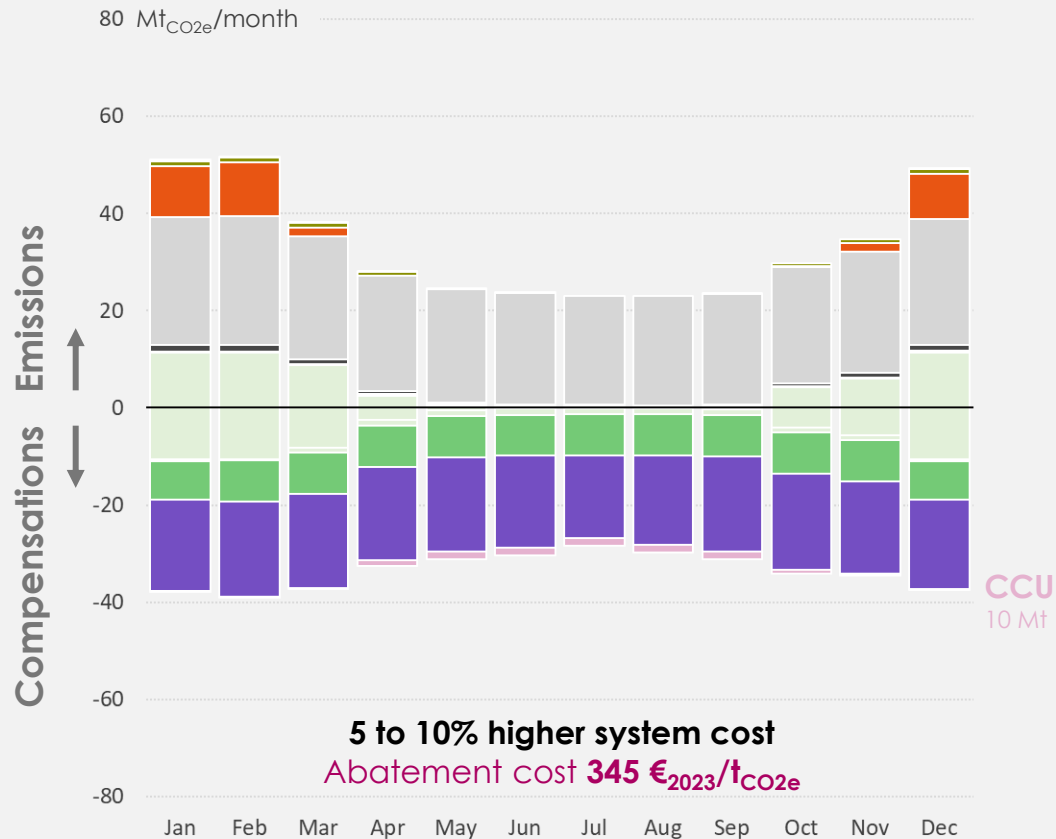
10 countries together are carbon-neutral thanks to production of bioenergy, import of green molecules, and massive deployment of carbon capture



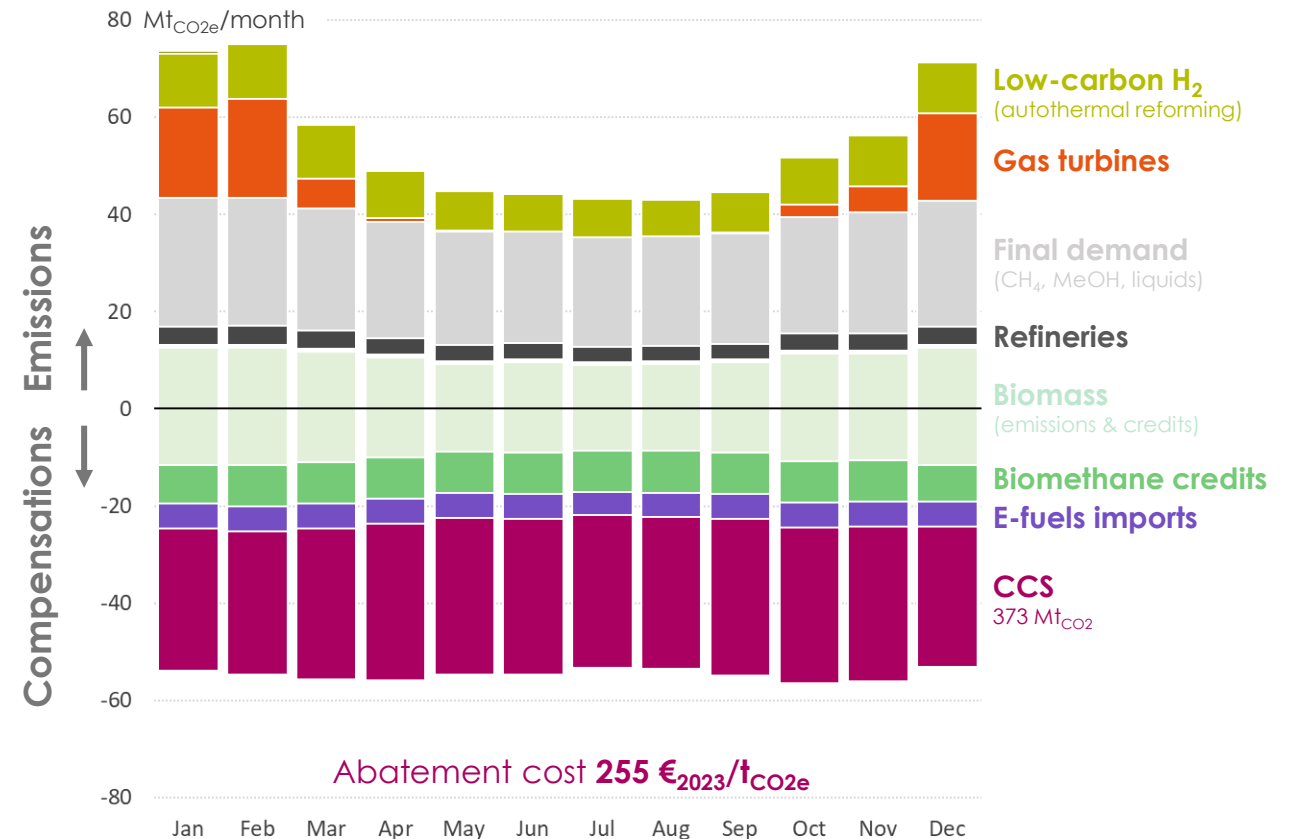
7. Carbon capture is a no-brainer (2/2)

Not allowing CCS would increase system cost (by 5-10%) and import dependency from overseas (for synthetic molecules)

No CCS

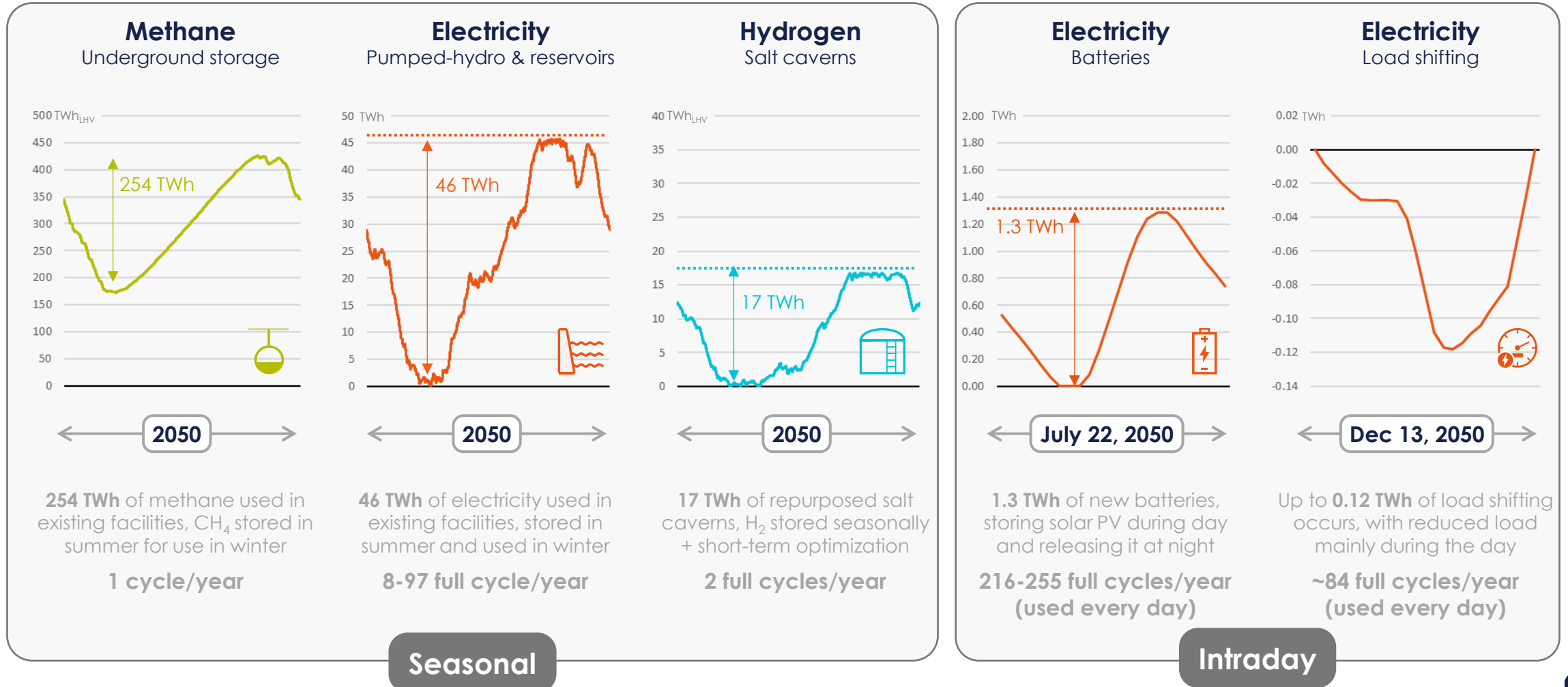


CO₂ air balance (Distributed Energy scenario)



8. Molecule storage is essential for seasonal demand (1/2)

Seasonal storage is assured mainly with methane underground facilities, while intraday storage is covered with batteries and demand-side response

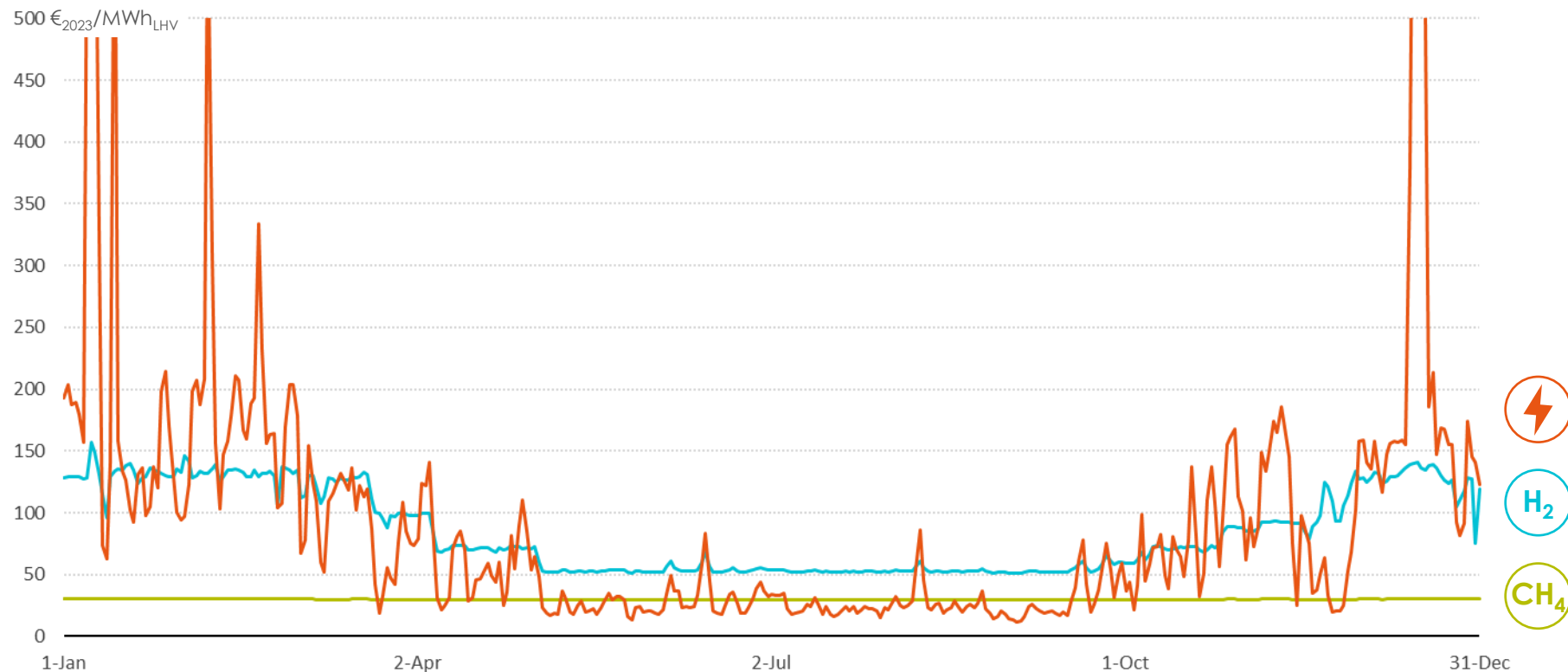


8. Molecule storage also helps stabilizing prices (2/2)

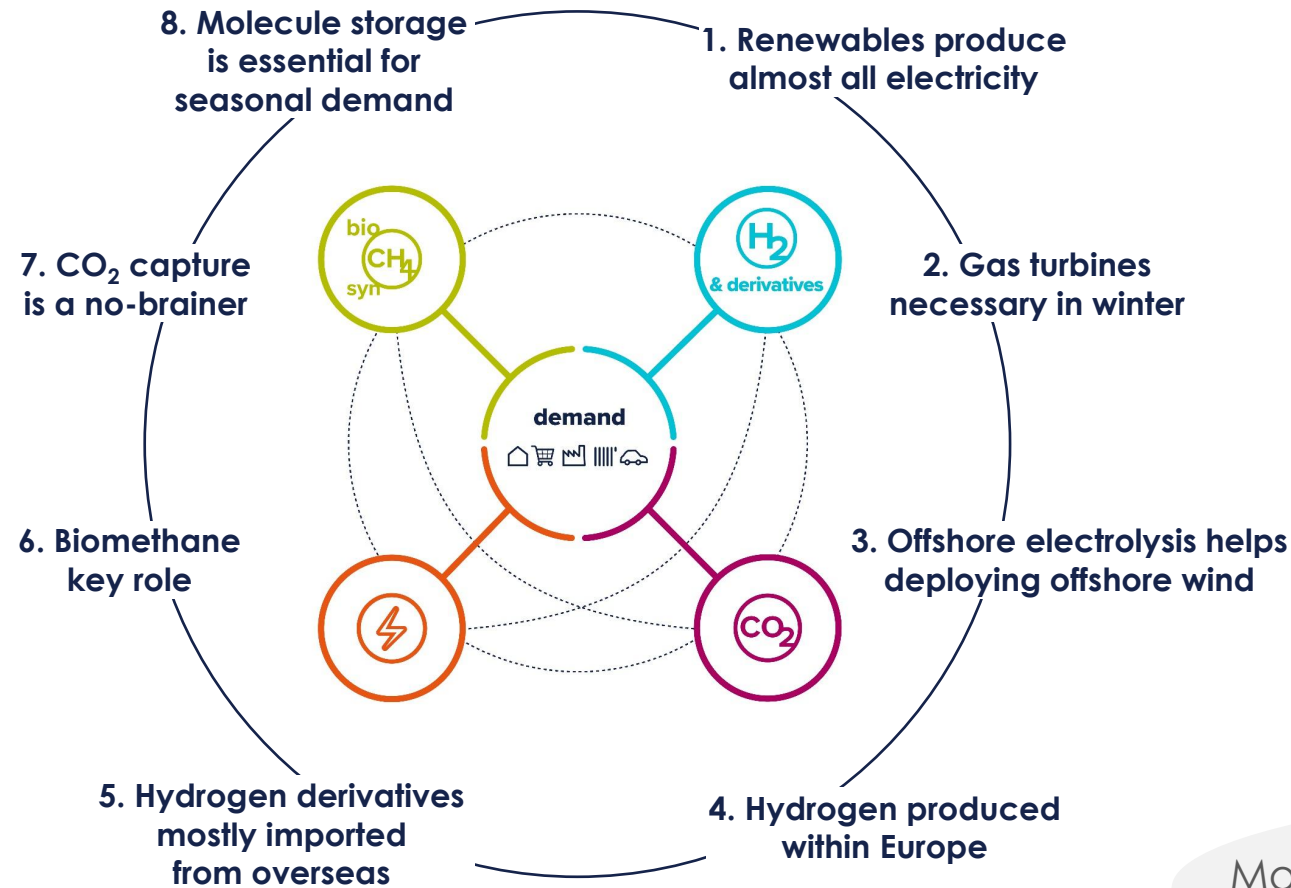
RES volatility can be seen in highly volatile marginal prices for electricity, while molecule prices are much more stable thanks to storage facilities



Marginal prices (Distributed Energy scenario)



Key insights from simulating ENTSOs TYNDP 2024 scenarios for 2050 with our North Sea Integration Model



A carbon-neutral energy system in 2050 in the North Sea countries is realistic and needs molecules

More details of our study can be found on [our website](#)

